

SCHEME OF EXAMINATION
&
DETAILED SYLLABUS

FOR

BACHELOR OF TECHNOLOGY (B.TECH/M.TECH) DUAL DEGREE
(4+2 Years)

FOR

INDUSTRIAL INTERNET OF THINGS (IIOT)

University School of Automation and Robotics
Batch of 2025-29
for 3rd - 8th Sem
as per NEP (2020)



University School of Automation and Robotics

GURU GOBIND SINGH INDRAPRASTHA UNIVERSITY
EAST DELHI CAMPUS, SURAJMAL VIHAR-110092

Knowledge and Attitude Profile (WK)

- **WK1:** A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.
- **WK2:** Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
- **WK3:** A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
- **WK4:** Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
- **WK5:** Knowledge, including efficient resource use, environmental impacts, whole-life cost, reuse of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.
- **WK6:** Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.
- **WK7:** Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.
- **WK8:** Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.
- **WK9:** Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.

Programme Outcomes

- **PO1: Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in **WK1 to WK4** respectively to develop solutions for complex engineering problems.
- **PO2: Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (**WK1 to WK4**)
- **PO3: Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for public health and safety, whole-life cost, net zero carbon, culture, society, and the environment as required. (**WK5**)
- **PO4: Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis and interpretation of data to provide valid conclusions. (**WK8**)
- **PO5: Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling, recognizing their limitations to solve complex engineering problems. (**WK2 and WK6**)
- **PO6: The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for their impact on sustainability with reference to economy, health, safety, legal framework, culture, and the environment. (**WK1, WK5, and WK7**)
- **PO7: Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national and international laws. (**WK9**)
- **PO8: Individual and Collaborative Team Work:** Function effectively as an individual, and as a member or leader in diverse and multidisciplinary teams.
- **PO9: Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.
- **PO10: Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects in multidisciplinary environments.
- **PO11: Life-Long Learning:**
Recognize the need for, and have the preparation and ability for:
 - i) independent and life-long learning;
 - ii) adaptability to new and emerging technologies; and
 - iii) critical thinking in the broadest context of technological change. (**WK8**)

Course / Paper Group Codes:

DSC: Discipline Specific Course and **ES** (Engineering Science) should be considered as part of DSC

MSC: Minor Specific Course

VAC: Value Added Course

MDC: Multi Disciplinary Course

AEC: Ability Enhancement Course

SEC: Skill Enhancement Course

SI: Summer Internship

Definitions:

Batch: The batch of the student shall mean the year of the first time enrolment of the students in the programme of study in the first semester. Lateral entry students admitted in the 3rd semester / 2nd year shall be designated as students admitted in the previous batch as they are admitted one year later. A student re-admitted in a programme of study in a lower/later batch shall be considered as the student of the original batch for calculation of the duration of the study.

Programme of study shall mean Bachelor of Technology.

Acronyms:

APC: Academic programme committee comprising all faculty of the school.

L: Number of Lecture hours per week

T/P: Number of Tutorial / Practical Hours per week

C: Number of credits assigned to a course / paper

COE: Controller of Examinations of the Examinations Division of the University.

SGPA/CGPA: Semester/Cumulative Grade Point Average.

NUES: No end term examination shall be held. The evaluation shall be conducted as per the scheme of examinations as described in the scheme of study.

Third Semester

Group	Paper Code	Paper Title	L	P	Credits
Theory Papers					
DSC	ARI201	Computer Networks	3	-	3
DSC	ARI203	Digital Electronics	3	-	3
MSC	ARI205	Sensing Technologies for IoT	2	-	2
MSC	ARI207	Design and Analysis of Algorithms	3	-	3
MDC	ARI209	Probability and Statistics	3	-	3
AEC	ARI211	Technical Writing ^{*(NUES)}	2	-	2
Practical/Viva-Voce					
DSC	ARI251	Computer Networks Lab	-	2	1
DSC	ARI253	Digital Electronics Lab	-	2	1
SEC	ARI255	Sensors and Instrumentation Lab	-	2	1
MSC	ARI257	Design and Analysis of Algorithms Lab	-	2	1
VAC	ARI259	Database Management Systems Lab	-	2	1
VAC	ARI261	OOPS with JAVA Lab	-	2	1
					22

Fourth Semester

Group	Paper Code	Paper Title	L	P	Credits
Theory Papers					
DSC	ARI202	Internet of Things Architecture and Protocol	3	-	3
SEC	ARI204	PLC-SCADA	3	-	3
DSC	ARI206	Signals and Systems	3	-	3
MDC	ARI208	Artificial Intelligence & Machine Learning for IIOT	3	-	3
MSC	ARI210	Control Systems	3	-	3
Practical/Viva-Voce					
DSC	ARI252	Internet of Things Architecture and Protocols Lab	-	2	1
SEC	ARI254	PLC-SCADA Lab	-	2	1
DSC	ARI256	Signals and Systems Lab	-	2	1
MDC	ARI258	Artificial Intelligence & Machine Learning for IIOT Lab	-	2	1
MSC	ARI260	Control Systems Lab	-	2	1
AEC	ARI262	Data Visualization Lab	-	2	1
AEC	ARI264	PCB Design Lab	-	2	1
					22

(NUES): Non-University Exam Subject, Comprehensive evaluation by the concerned teacher, out of 100, as per detailed syllabus.

Students are supposed to enroll for Summer Internship (SI w.r.t. ARI361) upon completion of 4th semester and whose evaluation will be performed towards the end of 5th Semester.

Fifth Semester

Approved by BoS of USAR : 16/6/2026

Approved by AC sub-committee : 20/07/2026

Applicable for Batch admitted in 2025 and Lateral Entry of AS 2026-27 admitted in Batch 2025

Group	Paper Code	Paper Title	L	P	Credits
Theory Papers					
SEC	ARI301	Electronic Design Automation for VLSI	3	-	3
MDC	ARI303	Analog and Digital Communication Systems	3	-	3
DSC	ARI305	Wireless Sensor Networks	3	-	3
MSC	As per the MSC List	One MSC from the MSC List as per the decision of the APC (Academic Program Committee) of the School (MSC-I)	3	-	3
MSC	As per the MSC List	One MSC from the MSC List as per the decision of the APC (Academic Program Committee) of the School (MSC-II)	3	-	3
Practical/Viva-Voce					
SEC	ARI351	Electronic Design Automation for VLSI Lab	-	2	1
MDC	ARI353	Analog and Digital Communication Systems Lab (MATLAB/Labview)	-	2	1
DSC	ARI355	Wireless Sensor Networks Lab	-	2	1
MSC	As per the MSC List	MSC-I Lab	-	2	1
MSC	As per the MSC List	MSC-II Lab	-	2	1
SI	ARI361	Summer Internship-I ^{## (NUES)}	-	-	2
					22

(NUES): Non-University Exam Subject, Comprehensive evaluation by the concerned teacher, out of 100, as per detailed syllabus.

Sixth Semester

Group	Paper Code	Paper Title	L	P	Credits
Theory Papers					
DSC	ARI302	Embedded Systems	3	-	3
AEC	ARI304	Data Analytics for IoT	3	-	3
DSC	ARI306	Cyber Security and Cryptography for Industrial Systems	3	-	3
MSC	ARI308	Cloud, Dew, Edge and Fog Computing	3	-	3
MSC	As per the MSC List	One MSC from the MSC List as per the decision of the APC (Academic Program Committee) of the School (MSC-III)	3	-	3
MSC	As per the MSC List	One MSC from the MSC List as per the decision of the APC (Academic Program Committee) of the School (MSC-IV)	3	-	3
Practical/Viva-Voce					
DSC	ARI354	Embedded Systems Lab	-	2	1
AEC	ARI356	Data Analytics for IoT Lab	-	2	1
MSC	ARI358	Cloud, Dew, Edge and Fog Computing Lab	-	2	1
MSC	As per the MSC List	MSC-III Lab	-	2	1
					22
NCC/NSS//DSW Clubs/Tech Clubs/Societies of USAR	ARI352	NCC/NSS/ Club/ DSW Clubs / USAR Tech and Non Tech Clubs (NUES)			2

Students are supposed to enroll for Summer Internship (SI w.r.t. ARI465) upon completion of 6th semester and whose evaluation will be performed towards the end of 7th Semester

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Seventh Semester

Theory Papers					
Group	Paper Code	Paper Title	L	P	Credits
DSC	ARI401	FPGA Based System Design	3	-	3
DSC	ARI403	Industrial Photonics	3	-	3
DSC	ARI405	Real Time Operating System	3	-	3
MSC	As per the MSC List	One MSC from the MSC List as per the decision of the APC (Academic Program Committee) of the School (MSC-V)	3	-	3
MSC	As per the MSC List	One MSC from the MSC List as per the decision of the APC (Academic Program Committee) of the School (MSC-VI)	3	-	3
Practical/Viva-Voce					
DSC	ARI451	FPGA System Design using HDL Lab	-	2	1
MSC	As per the MSC List	MSC VI Lab	-	2	1
SI	ARI465	Summer Internship ^{##} (NUES)	-	-	2
DSC	ARI467	Minor Project ^{**}	-	-	3
					22

Comprehensive evaluation by a committee of teachers, constituted by the Academic Programme Committee (APC), out of 100. The training shall be of 4 to 6 weeks duration. The training can be under the mentorship of a teacher of the school.

*** The student shall be allocated a supervisor/guide for project work at the start of 7th semester by the school, preferably, the project can be continued into the 8th semester. In the 7th semester evaluation, the criteria for evaluation shall be the conceptualization of the project work, the background study/literature survey and the identification of objectives and methodology to be followed for the project. In the absence of the supervisor, the Dean of the school can assign the responsibility of the supervisor (for the purpose of examinations) to any faculty of the school. The internal and external bifurcation of the project marks will be as per the bifurcation of marks for the practical examination.*

Students are supposed to choose relevant topic for the Research Seminar (ARI402). The mentor for the same would be assigned in 7th Semester and its evaluation will be performed in 8th Semester. Students are encouraged to go for Research publication in reputed journal/conference.

Eight Semester

Group	Paper Code	Paper Title	L	P	Credits
Practical/Viva-Voce					
DSC	ARI402	Research Seminar####(NUES)	-	-	4
DSC	ARI452/ ARI454	Major Project*** / Industry Internship#####	-	-	16
					20

***The student shall be allocated a supervisor/guide for project work at the start of the semester by the school. The criteria for evaluation shall be the conceptualization of the project work, the background study/literature survey and the identification of objectives and methodology to be followed for the project. In the absence of the supervisor, the Dean of the school can assign the responsibility of the supervisor (for the purpose of examinations) to any faculty of the school. The internal and external bifurcation of the project marks will be as per the bifurcation of marks for the practical examination.

Students have the option to pursue his/her Dissertation on the basis of the Live Projects in a Recognized (CIN No. Required) Company/ Organization. The proposed company/ organization must be approved by the Dean/APC.

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SEMESTER WISE LIST OF MINOR SPECIALIZATION COURSE [MSC]

1. MSC shall be offered in various semesters as per the scheme of the program.
2. MSC shall be offered if at least 1/3rd of the total program strength opts for the course.
3. MSC subjects can be changed by APC in due course of time.

Category	Paper Code		Course Name	L	P	Credit s
MSC - I, II (Semester 5: choose any course)						
MSC-I	ARI371	ARI371T	Deep Learning	3	-	3
		ARI371P	Deep Learning Lab	-	2	1
	ARI373	ARI373T	IoT enabled smart grid and energy systems	3	-	3
		ARI373P	IoT enabled smart grid and energy systems Lab	-	2	1
MSC-II	ARI375	ARI375T	Mobile Application Development for IoT	3	-	3
		ARI375P	Mobile Application Development for IoT Lab	-	2	1
	ARI377	ARI377T	Electrical vehicles and drives	3	-	3
		ARI377P	Electrical vehicles and drives Lab	-	2	1
MSC - III, IV (Semester 6: choose any course)						
MSC-III	ARI372	ARI372T	Fundamentals of Robotics	3	-	3
		ARI372P	Robotics Lab	-	2	1
	ARI374	ARI374T	Introduction to unmanned Systems	3	-	3
		ARI374P	Introduction to unmanned Systems Lab	-	2	1
MSC-IV	ARI376		Metaverse and Blockchain	3	-	3
	ARI378		Micro-electro-mechanical-systems (MEMS)	3	-	3
MSC - V, VI (Semester 7: choose any course)						
MSC-V	ARI471		Information Theory and Coding	3	-	3
	ARI473		System on chip	3	-	3
	ARI475		Brain computer interface	3	-	3
MSC-VI	ARI477	ARI477T	Image processing and computer vision	3	-	3
		ARI477P	Image processing and computer vision Lab	-	2	1
	ARI479	ARI479T	Generative Artificial Intelligence	3	-	3
		ARI479P	Generative Artificial Intelligence lab	-	2	1

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SEMESTER WISE LIST OF SKILL ENHANCEMENT COURSES (SEC)

Semester	Paper Code	Course Name	L	P	Credits
3 rd	ARI255	Sensors and Instrumentation Lab	-	2	1
4 th	ARI204	PLC-SCADA	3	-	3
	ARI254	PLC-SCADA Lab	-	2	1
5 th	ARI301	Electronic Design Automation for VLSI	3	-	3
	ARI351	Electronic Design Automation for VLSI Lab	-	2	1

SEMESTER WISE LIST OF VALUE-ADDED COURSES (VAC)

Semester	Paper Code	Course Name	L	P	Credits
3 rd	ARI259	Database Management Systems Lab	-	2	1
	ARI261	OOPS with JAVA Lab	-	2	1

SEMESTER WISE LIST OF ABILITY ENHANCEMENT COURSES (AEC)

Semester	Paper Code	Course Name	L	P	Credits
3 rd	ARI211	Technical Writing [#]	2	-	2
3 rd	ARI262	Data Visualization Lab	-	2	1
	ARI264	PCB Design Lab	-	2	1
6 th	ARI304	Data Analytics for IoT	3	-	3
	ARI356	Data Analytics for IoT Lab	-	2	1

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Important Notes:

1. a) The examinations, attendance criteria to appear in examinations, promotion and award of the degree shall be governed by the Ordinance 11 of the University.
b) Grading System shall be as per Ordinance 11 of the University.
2. Minimum duration of the Bachelor of Technology programme and Bachelor of Technology part of the Bachelor / Master of Technology (Dual Degree) programme shall be 4 years (N=4 years) (8 semesters) for the students admitted in the first year/ first semester of the degree programme.
3. Lateral entry students shall be admitted in the second year / third semester of the degree programme (will be considered effectively in the batch admitted in the first year in the previous academic session and shall be deemed to have been exempted from the courses/papers of the first year of the degree programme). No exemption certificate shall be issued in any case. The minimum exit time/year for lateral-entry students shall be the same as for the batch in which he/she is admitted as a lateral-entry student in the second year.
4. Maximum duration of the Bachelor of Technology programme shall be 7 years (N+3 years). After completion of N+3 years of study, no extension shall be given to the student for completing the requirements of the degree and the admission of the student shall stand cancelled.
5. (a) To earn an Honours degree, the student may enrol for 20 credits or more (over and above the total offered credits in the classroom) through SWAYAM / NPTEL MOOCs platform only. The acquisition of the credits should be completed before the 15th of July of the graduating year.
(b) Honours degree shall be awarded if and only if at least 20 credits are acquired through MOOCs. To obtain Honours in the programme, the student must apply for the same to the school/institute/college in the fifth semester by 30th August. The school/institute/college shall inform the list of such students to the Examination Division of the University.
(c) The specific courses through MOOCs shall be registered by the student only after approval of a sub-committee of the Academic Programme Committee (APC) approved by the Dean/Director/Principal of the school/institute/college. The APC shall approve the course if it is not already studied by the student or the student shall not study it in future and adds value to the major area of specialization of the degree. The result of the approved papers shall be submitted by the student to the school/institute/college to be forwarded to the Examination Division of the University.
(d) The cost of taking the MOOC course is to be borne by the concerned student. Such courses shall be reflected as additional courses/papers for the student.
(e) The papers through MOOCs for the Honours Degree shall not be part of the curriculum considered for the computation of the student's SGPA/CGPA.
(f) No Honours shall be conferred if the degree requirements are not completed within the minimum duration.
(g) If a student has qualified any subject, which is a component of another online/offline degree program offered by any other university in India or abroad, it shall not be counted towards the award of Honours degree.
(h) No Honours shall be conferred if the student fails in any of the papers offered to the student, at the time of verification for the award of the honours degree. That is, the Honours degree shall be conferred only to students with full credits of total offered credits.
(i) No Honours shall be conferred if the student has taken any of the interim degrees (as per MEME guidelines of University).
(j) The Honours degree shall only be awarded if the CGPA of the student is 7.5 and above. In addition,

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the student should be eligible for the award of the degree after the immediate completion of the fourth year of the course.

6. Multiple Entry Multiple Exit (MEME) Option is as per guidelines of University (refer notification number: F.34 (11)(13)/2025/DAA/1448 dated:24.03.2026).
7. Credit Summary:
 - a. The students shall undergo the group of Courses / Papers (Refer Table 1) as enumerated in the scheme (For the students admitted in the First Year / First Semester)

Table 1: Distribution of Credits

Year	1		2		3		4		Total Credits
Semester	1	2	3	4	5	6	7	8	
DSC	16	16	8	8	4	7	13	20	92
MSC			6	4	8	11	7		36
MDC		3	3	4	4				14
AEC	2		2	2		4			10
SEC		1	1	4	4				10
VAC	4	2	2						8
SI					2		2		4
Total	22	22	22	22	22	22	22	20	174**
	44		44		44		42		
Minimum Requirement	40 [#]		80 ^{##}		120 ^{###}		160 [*]		

*Minor Project, Major Project, Summer Internship (SI), and Research Seminar are mandatory components of the programme. Students are not permitted to drop, seek exemption from, or leave any of these courses in any semester. NSS/NCC/DSW Clubs/Tech or Non-Tech Clubs/Societies of school/institute/college worth 2 credits to be earned over and above the minimum requirements.

For **B.Tech Honours, MOOCS of 20 credits are to be earned over and above the Total Credits offered.

Students intending to exit the programme after the completion of the first year as per MEME of university will obtain **Certificate in IIOT**, subject to earning 04 additional credits (over and above the minimum credit requirement prescribed) by doing domain-specific internship/ apprenticeship of minimum 8 weeks either in the University or in Industry that would help the candidates to acquire job-ready competencies required to enter the workforce.

Students intending to exit the programme after the completion of the second year as per MEME of university will obtain **Diploma in IIOT**, subject to earning 04 additional credits (over and above the minimum credit requirement prescribed) by doing domain-

specific internship/ apprenticeship of minimum 8 weeks either in the University or in Industry that would help the candidates to acquire job-ready competencies required to enter the workforce. The student upon rejoining for third year may not require to apply for additional summer internship (if any such course requirement exists for fifth semester), his internship to meet exit condition as per prescribed guideline of MEME, may be considered for evaluation of such course.

Students intending to exit the programme after the completion of the third year as per MEME of university will get **Bachelor of Science (BSc) in IIOT**, subject to fulfillment of minimum requirement as prescribed in MEME of university.

b. The students admitted as Lateral Entry shall undergo Courses / Papers as enumerated in the scheme after third semester need to earn minimum 120* credits for completion of degree (i.e., first year is exempted from Table 1) and guidelines of MEME of University shall be followed for any interim degree. For B.Tech Honours, MOOCS of 20 credits are to be earned over and above the Total Credits offered.

8. Only after qualifying for the award of the degree of Bachelor of Technology, the student may be allowed to proceed to the Master in Technology part of the Bachelor / Master of Technology (Dual Degree)[%].
9. The scheme and syllabi of the Master of Technology part of the Bachelor / Master of Technology (Dual Degree) is notified separately[%].

NOTE: [%] Points 8 and 9 apply to USS-USAR only.

DETAILED
SYLLABUS
FOR
1st SEMESTER
And
2nd SEMESTER

Refer to the document named “B.Tech (IIOT) AS 2025-26” available in “Scheme/Syllabi of All First Year B.Tech Programs of USAR” section on the university website (EDC).

DETAILED SYLLABUS FOR 3rd SEMESTER

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Paper Code: ARI201								L	P	Credit	
Paper Title: Computer Networks								3	0	3	
Marking Scheme: Teachers Continuous Evaluation: As per university examination norms from time to time. End Term Theory Examination: As per university examination norms from time to time.											
INSTRUCTIONS TO PAPER SETTERS: Maximum Marks : As per University norms											
➤ There should be 9 questions in the end term examination question paper ➤ Question No. 1 should be compulsory and cover the entire syllabus. This question should have objective or short answer type questions. ➤ Apart from Question No. 1, the rest of the paper shall consist of four units As per the syllabus. Every unit should have two questions. However, students may be asked to attempt only 1 question from each unit. ➤ The questions are to be framed keeping in view the learning outcomes of course/paper. The standard/ level of the questions to be asked should be at the level of the prescribed textbooks. ➤ The requirement of (scientific) calculators/ log-tables/ data-tables may be specified if required											
Course Outcomes [Bloom’s Knowledge Level (KL)]: CO1: To Understand the basic concepts of Computer Network. [K1, K2] CO2: Apply data link layer concepts including framing, error detection, flow control mechanisms, MAC protocols, and Ethernet standards in network design and analysis. [K1, K2, K3] CO3: To Analyze the different Routing Techniques and IP Addressing Schemes and wireless/ad hoc routing protocols such as AODV. [K4] CO4: Evaluate transport and application layer protocols, including TCP, UDP, socket programming concepts, DNS, HTTP, FTP, SMTP, and electronic mail architecture for modern computer networking applications. [K3, K4].											
CO/PO	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11
CO1	3	3	3	3	1	1	-	-	-	-	2
CO2	3	3	3	3	1	2	-	-	-	-	2
CO3	3	3	3	3	1	2	-	-	-	-	3
CO4	3	3	3	3	1	2	-	-	-	-	3
Course Content											No of Lectures
Unit-I Introduction to data communication; concept of layering; OSI reference model and TCP/IP protocol stack; comparison of OSI and TCP/IP models. Data and signals; digital and analog transmission techniques. Bandwidth utilization techniques. Switching techniques: circuit switching, packet switching and virtual circuit switching.											[8]
Unit-II Data Link Layer: Functions of the data link layer; Framing techniques; Error detection and Correction Techniques; Introduction to flow control, stop and wait protocol, sliding window protocol. Medium Access Control (MAC) protocols. Ethernet (IEEE 802.3) standards and Ethernet bridging.											[8]

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Unit-III Network Layer: Functions of the network layer; routing algorithms and protocols including shortest path routing, flooding, distance vector routing and link state routing. Fragmentation. IP addressing; IPv4 addressing and CIDR notation. IP support protocols: ARP, DHCP and ICMP. Network Address Translation (NAT). Introduction to Wireless Networks: Wireless LAN (IEEE 802.11) overview, characteristics of wireless networks, challenges in wireless routing. Ad hoc Routing Protocol: Ad hoc On-Demand Distance Vector (AODV) routing protocol – concept, route discovery, route maintenance, advantages and limitations.	[8]
Unit-IV Transport Layer: Transport layer services: flow control and congestion control; User Datagram Protocol (UDP); Transmission Control Protocol (TCP); Multiplexing and Demultiplexing, socket programming concepts. Application Layer: Application layer protocols: Domain Name System (DNS), Hypertext Transfer Protocol (HTTP), File Transfer Protocol (FTP), Simple Mail Transfer Protocol (SMTP); electronic mail architecture.	[8]
Text Books [T1] Forouzan, B. A. (2017). Data communications and networking (5th ed.). McGraw-Hill Education. [T2] Kurose, J. F., & Ross, K. W. (2021). Computer networking: A top-down approach (8th ed.). Pearson Education. [T3] Tanenbaum, A. S., & Wetherall, D. J. (2021). Computer networks (6th ed.). Pearson Education.	
Reference Books [R1] Peterson, L. L., & Davie, B. S. (2022). Computer networks: A systems approach (6th ed.). Morgan Kaufmann. [R2] Comer, D. E. (2019). Internetworking with TCP/IP: Principles, protocols, and architecture (Vol. 1, 6th ed.). Pearson Education. [R3] Stallings, W. (2014). Data and computer communications (10th ed.). Pearson Education. [R4] Lammle, T. (2020). CCNA 200-301 official cert guide, volume 1 (1st ed.). Wiley.	

Paper code: ARI203								L	P	Credit	
Paper Title: Digital Electronics								3	0	3	
Marking Scheme: Teachers Continuous Evaluation: As per university examination norms from time to time. End Term Theory Examination: As per university examination norms from time to time.											
INSTRUCTIONS TO PAPER SETTERS: Maximum Marks: As per University norms											
➤ There should be 9 questions in the end term examination question paper ➤ Question No. 1 should be compulsory and cover the entire syllabus. This question should have objective or short answer type questions. ➤ Apart from Question No. 1, the rest of the paper shall consist of four units As per the syllabus. Every unit should have two questions. However, students may be asked to attempt only 1 question from each unit. ➤ The questions are to be framed keeping in view the learning outcomes of course/paper. The standard/ level of the questions to be asked should be at the level of the prescribed textbooks. ➤ The requirement of (scientific) calculators/ log-tables/ data-tables may be specified if required											
Course Outcomes [Bloom’s Knowledge Level (KL)]: CO1: Understand number systems and its applications. [K1, K2] CO2: Boolean expressions minimization, logic gates and its applications to digital circuits design, Combinational circuits using logic gates. [K1, K3, K4] CO3: Design Sequential logic Circuits and its application in synchronous and asynchronous counter circuit design. [K1, K3, K4] CO4: Ability of students to understand the basic concepts of digital logic Families. [K1, K2]											
CO/PO	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11
CO1	3	2	1	1	-	-	-	-	-	-	1
CO2	1	2	3	-	-	-	-	-	-	-	2
CO3	1	2	2	-	-	-	-	-	-	-	2
CO4	3	2	1	1	-	-	-	-	-	-	1
Course Content											No of Lectures
Unit-I Number system - Decimal, Binary, Octal and Hexadecimal – conversion from one system to another – representation of negative numbers – representation of BCD numbers – character representation – character coding schemes – ASCII – EBCDIC, Excess-3 and Gray code. Addition, subtraction, multiplication and division of binary numbers. Addition and subtraction of BCD, Octal and Hexadecimal numbers.											[8]

Unit-II Introduction — Postulates of Boolean algebra – Canonical and Standard Forms — logic functions and logic gates, methods of minimization of logic functions — Karnaugh map method and Quin McClusky method Product-of-Sums Simplification — Don't-Care Conditions. Combinational Logic: Analysis and design procedure, Half and Full Adders, Subtractors, 4-bit BCD adder/subtractor, Magnitude comparator, Multiplexers, Demultiplexers, Decoders, Encoders, Parity Generator/Checker.	[8]
Unit-III Sequential Logic and Its Applications: Basic latches & Flip Flops, Characteristic Equations of Flip Flops, Flip Flop Conversion, Master Slave Flip Flop, Shift Registers: SISO, SIPO, PISO, PIPO. Synchronous & Asynchronous Sequential Circuits: Design of Up counter, Down Counter, Up/Down Counter, Ripple Counters, MOD counter, Johnson, Ring Counter & BCD counter.	[8]
Unit-IV Memory & Programmable Logic Devices: Digital Logic Families: TTL, ECL, CMOS Logic families, Fan in, Fan Out, power dissipation, Figure of Merit, Noise Margin; RAM, ROM, Programmable Logic Arrays (PLA), Programmable Array Logic (PAL).	[8]
Text Books [T1] Mano, M. M. (2013). Digital Logic and Computer Design (4th ed.). Pearson Education. [T2] Floyd, T. L. (2020). Digital Fundamentals (11th ed.). Pearson [T3] Harris, D. M., & Harris, S. L. (2021). Digital Design and Computer Architecture: RISC-V Edition (1st ed.). Morgan Kaufmann.	
Reference Books [R1] Tokheim, R. L. (2014). Digital Electronics: Principles and Applications (8th ed.). McGraw Hill Education. [R2] Rajaraman, V., & Radhakrishnan, T. (2012). An Introduction to Digital Computer Design (5th ed.). PHI Learning Pvt. Ltd. [R3] Leach, D. P., Malvino, A. P., & Saha, G. (2015). Digital Principles and Applications (8th ed.). McGraw Hill Education.	

Paper code: ARI205							L	P	Credit		
Paper Title: Sensing Technologies for IoT							2	0	2		
Marking Scheme:											
Teachers Continuous Evaluation: As per university examination norms from time to time.											
End Term Theory Examination: As per university examination norms from time to time.											
INSTRUCTIONS TO PAPER SETTERS: Maximum Marks: As per University norms											
➤ There should be 9 questions in the end term examination question paper ➤ Question No. 1 should be compulsory and cover the entire syllabus. This question should have objective or short answer type questions. ➤ Apart from Question No. 1, the rest of the paper shall consist of four units As per the syllabus. Every unit should have two questions. However, students may be asked to attempt only 1 question from each unit. ➤ The questions are to be framed keeping in view the learning outcomes of course/paper. The standard/ level of the questions to be asked should be at the level of the prescribed textbooks. ➤ The requirement of (scientific) calculators/ log-tables/ data-tables may be specified if required.											
Course Outcomes [Bloom’s Knowledge Level (KL)]:											
CO1: Ability of students to understand the principle of sensors and their characteristics. [K1, K2]											
CO2: Ability of students to develop basic understanding of Smart Sensors, Signal Conditioning and ADC [K1, K2, K3]											
CO3: Ability of students to understand concepts of Physical and Environmental Sensing. [K1, K2]											
CO4: Ability of students to understand Motion, Force, and Spatial Sensing. [K1, K2]											
CO/PO	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11
CO1	3	3	3	3	1	-	-	-	-	-	2
CO2	3	3	3	3	1	2	-	-	-	-	2
CO3	3	3	3	3	1	2	-	-	1	-	3
CO4	3	3	3	3	1	2	-	-	1	-	2
Course Content										No of Lectures	
Unit-I Sensor Classification and Characteristics: Fundamental Definitions: Transducers, sensors, and actuators; smart sensors vs. conventional sensors. Static Characteristics: Accuracy, precision, sensitivity, resolution, linearity, hysteresis, and repeatability. Dynamic Characteristics: Zero-order, first-order, and second-order sensor responses; time constant and frequency response.										[4]	
Unit-II Smart Sensor Architecture: Intelligent Instrumentation: Components that characterize an intelligent sensor (self-calibration, self-diagnosis, and multi-sensing). Standardization: IEEE 1451 standards for smart sensors and networking, Fieldbus and interfaces. Signal Conditioning and Analog-to-Digital Conversion (ADC): Signal Amplification, Instrumentation amplifiers and operational amplifier (Op-Amp) circuits for weak sensor signals. Filtering: Low-pass, high-pass, and band-pass analog filters for noise reduction. Quantization &										[6]	

Approved by BoS of USAR : 16/6/2026

Approved by AC sub-committee : 20/07/2026

Applicable for Batch admitted in 2025 and Lateral Entry of AS 2026-27 admitted in Batch 2025

Sampling: ADC architectures (Successive Approximation, Flash, Sigma-Delta), resolution (8-bit vs. 12-bit vs. 16-bit), and the Nyquist sampling theorem.	
Unit-III Physical and Environmental Sensing: Thermal Sensors: Thermistors (NTC/PTC), RTDs (Pt100), thermocouples, and digital IC sensors (LM35, DS18B20). Humidity and Moisture: Capacitive and resistive humidity sensing; soil moisture probes (DHT11/DHT22 mechanics). Optical & Proximity Sensors: Photodiodes, LDRs (Light Dependent Resistors), infrared (IR) sensors, and ultrasonic distance sensors (HC-SR04).	[5]
Unit-IV Motion, Force, and Spatial Sensing: Inertial Measurement Units (IMUs): MEMS accelerometers (acceleration/tilt) and gyroscopes (angular velocity). Mechanical Sensors: Strain gauges, load cells for weight measurement, and piezoelectric pressure sensors. Acoustic & Gas Sensors: Condenser microphones, MQ-series electrochemical gas sensors (MQ-2, MQ-135 for air quality).	[5]
Text Books [T1] Khanna, V. K. (2024). IoT Sensors: An Exploration of Sensors for Internet of Things (1st ed.). CRC Press. [T2] Mukhopadhyay, S. C., Jayasundera, K. P., & Postolache, O. A. (Eds.). (2019). Modern Sensing Technologies (1st ed.). Springer International Publishing.	
Reference Books [R1] Gupta, G. S. (Ed.). (2008). Smart sensors and sensing technology (1st ed.). Springer Science & Business Media. [R2] Misra, S., Mukherjee, A., & Roy, A. (2021). Introduction to IoT (1st ed.). Cambridge University Press.	

Paper code: ARI207								L	P	Credit	
Paper Title: Design and Analysis of Algorithms								3	0	3	
Marking Scheme: Teachers Continuous Evaluation: As per university examination norms from time to time. End Term Theory Examination: As per university examination norms from time to time.											
INSTRUCTIONS TO PAPER SETTERS: Maximum Marks: As per University norms											
➤ There should be 9 questions in the end term examination question paper. ➤ Question No. 1 should be compulsory and cover the entire syllabus. This question should have objective or short answer type questions. ➤ Apart from Question No. 1, the rest of the paper shall consist of four units As per the syllabus. Every unit should have two questions. However, students may be asked to attempt only 1 question from each unit. ➤ The questions are to be framed keeping in view the learning outcomes of course/paper. The standard/ level of the questions to be asked should be at the level of the prescribed textbooks. ➤ The requirement of (scientific) calculators/ log-tables/ data-tables may be specified if required.											
Course Outcomes [Bloom’s Knowledge Level (KL)]: CO1: Analyze the performance of algorithms using asymptotic notations, recurrence relations, and complexity measure to evaluate computational efficiency. [K4] CO2: Apply Divide and Conquer and Dynamic Programming techniques to design efficient solutions for optimization and computational problems. [K3] CO3: Design and analyze Greedy and Graph algorithms for solving optimization and network-related problems. [K4] CO4: Evaluate computational complexity classes and distinguish between tractable and intractable problems using NP-completeness concepts. [K5]											
CO/PO	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11
CO1	3	3	1	2	1	-	-	-	-	-	2
CO2	3	3	3	2	1	-	-	-	-	-	3
CO3	3	3	3	2	1	-	-	-	-	-	3
CO4	3	3	2	3	-	-	-	-	-	-	2
Course Content											No of Lectures
Unit-I Foundations of Algorithms and Complexity Analysis: Overview of fundamental data structures (Linear Data Structures, Non-Linear Data Structures (Trees and Graphs)), Characteristics of Algorithms and Pseudocode, Time and Space Complexity Analysis, Asymptotic Notations: Big-O, Big-Ω, Θ, Little-o, Little-ω, Growth of Functions, Recurrence Relations- Iteration Method, Recursion Tree Method, Substitution Method, Master Method.											[8]

Unit-II Divide and Conquer and Dynamic Programming: Divide and Conquer-General Methodology, Merge Sort, Quick Sort, Strassen's Matrix Multiplication, Dynamic Programming- Elements of Dynamic Programming, Matrix-chain multiplication, Longest common subsequence, Optimal binary search trees, 0-1 Knapsack Problem.	[8]
Unit-III Greedy Algorithms and Graph Algorithms: Greedy Strategy-Elements of Greedy Method, Local and Global Optima, Applications-Activity Selection Problem, Fractional Knapsack Problem, Task Scheduling Problem. Graph Algorithms-Minimum Spanning Trees: Kruskal's Algorithm, Prim's Algorithm, Single Source Shortest Paths: Dijkstra's Algorithm, Bellman-Ford Algorithm, Floyd-Warshall Algorithm.	[8]
Unit-IV Computational Complexity and NP-Completeness: Tractable and Intractable Problems, Class P and NP, Polynomial-Time Verification, Reducibility, NP-Completeness, NP-Hard Problems, Case Studies: Vertex Cover Problem, Clique Problem.	[8]
Text Books [T1] Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C. (2022). Introduction to algorithms (4th ed.). MIT press. [T2] Kleinberg, J., & Tardos, É. (2022). Algorithm design (1st ed.). Pearson Education Limited.	
Reference Books [R1] Baase, S. (2009). Computer algorithms: introduction to design and analysis (3rd ed.). Pearson Education India. [R2] Skiena, S. S. (2020). The algorithm design manual (3rd ed.). New York: springer.	

Paper code: ARI209							L	P	Credit		
Paper Title: Probability and Statistics							3	0	3		
Marking Scheme: Teachers Continuous Evaluation: As per university examination norms from time to time. End Term Theory Examination: As per university examination norms from time to time.											
INSTRUCTIONS TO PAPER SETTERS: Maximum Marks: As per University norms											
➤ There should be 9 questions in the end term examination question paper. ➤ Question No. 1 should be compulsory and cover the entire syllabus. This question should have objective or short answer type questions. ➤ Apart from Question No. 1, the rest of the paper shall consist of four units As per the syllabus. Every unit should have two questions. However, students may be asked to attempt only 1 question from each unit. ➤ The questions are to be framed keeping in view the learning outcomes of course/paper. The standard/ level of the questions to be asked should be at the level of the prescribed textbooks. ➤ The requirement of (scientific) calculators/ log-tables/ data-tables may be specified if required.											
Course Outcomes [Bloom’s Knowledge Level (KL)]: CO1: Ability of students to understand basic probability concepts and random variables. [K1, K2, K3, K4] CO2: Ability of students to understand basic expectation, variance and discrete probability distributions used in engineering problems. [K2, K3, K4, K5] CO3: Ability of students to understand continuous probability distributions and joint distributions. [K1, K2, K3, K4, K5] CO4: Ability of students to learn confidence interval and hypothesis testing and apply in various engineering applications. [K1, K2, K3, K4, K5]											
CO/PO	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11
CO1	3	3	3	3	2	-	-	-	-	1	3
CO2	3	3	3	3	3	-	-	-	-	1	3
CO3	3	3	3	3	3	-	-	-	-	1	1
CO4	3	3	3	3	3	-	-	-	-	1	3
Course Content											No of Lectures
Unit-I Probability Theory: Basic concept of probability, Probability axioms, Sample space and Events, Independent and mutually exclusive events, Law of addition and multiplication of probabilities, Conditional probability, Law of total probability, Bayes’ theorem, Random variables: Discrete and continuous with their basic properties, probability mass function and probability distribution function, Cumulative distribution function, Conditional PDF.											[8]

Unit-II Statistics and Discrete Distributions: Independent random variables, Expectation and variance of random variables. Skewness and kurtosis, Covariance between random variables, Spearman's rank correlation coefficient, Moment generating function, Chebyshev's inequality, Discrete distributions: Discrete uniform, Bernoulli, Binomial, Poisson.	[8]
Unit-III Continuous distributions, Central limit theorem and Joint distributions: Continuous uniform, Normal, standard normal, exponential, t-distribution, chi-squared distribution. Sample mean, Central Limit Theorem, Sample variance, Joint probability mass function and joint probability density function, Expectation of two-dimensional random variables, Conditional mean and conditional variance.	[8]
Unit-IV Confidence Interval and Hypothesis testing: Confidence interval for population mean when variance is (i) known (ii) unknown, Confidence interval for variance of a normal distribution, Confidence interval for difference in means. Hypothesis testing: tests for means of normal distribution when variance is (i) known (ii) unknown. Testing the equality of means of two normal populations.	[8]
Text Books [T1] Ross, S. M. (2020). Introduction to probability and statistics for engineers and scientists (6th ed.). Academic press. [T2] Montgomery, D. C., & Runger, G. C. (2019). Applied statistics and probability for engineers (7th ed.). Wiley Publications. [T3] Gupta, S. C., and V. K. Kapoor (2020). Fundamentals of mathematical statistics (12th ed.). Sultan Chand & Sons Publications.	
Reference Books [R1] Hines, W. W., Montgomery, D. C., & Borror, D. M. G. C. M. (2008). Probability and statistics in engineering (4th ed.). John Wiley & Sons. [R2] Ravichandran J. (2023). Probability and Statistics for Engineers (1st ed.). Wiley. [R3] Miller I. and Miller M. (2013). Mathematical statistics with applications (8th ed.). Pearson India.	

Paper code: ARI211									L	P	Credit
Paper Title: Technical Writing [#] (NUES)									2	0	2
Marking Scheme: Teachers Continuous Evaluation: 100 marks This is an NUES paper, the examinations are to be conducted by the concerned teacher.											
Course Outcomes [Bloom’s Knowledge Level (KL)]: CO1: Ability to identify different types of Writings and apply evidence based Arguments. [K1,K2,K3,K5] CO2: Ability to interpret, list, assemble and formulate the writing process. [K1,K3, K6] CO3: Ability to remember, evaluate and design technical documents. [K1,K5,K6] CO4: Ability to classify and develop reference lists and research papers. [K2,K6]											
CO/PO	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11
CO1	2	2	2	2	2	-	-	3	3	-	1
CO2	1	3	1	1	2	-	-	3	3	-	1
CO3	3	1	3	1	3	-	-	3	3	-	1
CO4	1	1	1	3	1	-	-	3	3	-	3
Course Content											No of Lectures
Unit-I Writing Skills: Descriptive, Narrative, Argumentative and Discursive, Reflective and Literary-Evaluative Writing. Technical Writing: Definition, Purpose, Characteristics of Technical Writing.											[5]
Unit-II The Technical Writing Process: Prewriting Stage, Writing Stage and the Post-writing stage. Technical Writing Skills: Researching, Summarizing and Outlining, Visual Aids. Definition, Description, Set of Instructions.											[5]
Unit-III Formal Formatting: Arrangement of Formal Elements. Font Material. Format Devices in the Body of Formal Report-Heading, Pagination, End Material-Citations References and Bibliography. Appendix. In IEEE, APA style sheet and Latex.											[5]
Unit-IV Technical Writing Applications Memorandums, Formal and Informal Letter Format, Recommendations and Feasibility Reports. Proposals, Progress Reports. Analysis Reports and Job Applications, Presentation and Meetings.											[5]
Text Books [T1] Meenakshi, R., & Sharma, S. (2017). Technical Communication: Principles and Practice (2nd ed.). Oxford University Press, New Delhi. [T2] Forsyth, Sandy and Lesley Hutchison (1981), Practical Composition, Edinburgh: Oliver and Boyd.											
Reference Books [R1] Side, Charles H. (1999), How to Write and Present Technical Information (1st ed.). Cambridge University Press. [R2] Guffey, Mary Ellen (2000), Business Communication (4th ed.). Cincinnati: South-Western College Publishing.											

Web Page Link

Overleaf. (n.d.). Learn LaTeX in 30 minutes. Overleaf. Retrieved June 11, 2026, from

Paper code: ARI259								L	P	Credit	
Paper Title: Database Management System Lab								0	2	1	
Marking Scheme:											
Teachers Continuous Evaluation: As per university examination norms from time to time.											
End Term Theory Examination: As per university examination norms from time to time.											
Course Outcomes [Bloom's Knowledge Level (KL)]:											
CO1: Design and create relational databases using SQL and integrity constraints. [K2]											
CO2: Perform data manipulation and complex query processing using SQL. [K3]											
CO3: Develop PL/SQL programs using procedures, functions, cursors, and exception handling. [K4]											
CO4: Implement views, triggers, and advanced DBMS concepts for database management. [K5]											
CO/P O	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11
CO1	3	2	2	0	3	0	0	0	0	0	1
CO2	3	3	3	1	3	0	0	0	0	0	1
CO3	3	3	3	1	3	0	0	1	0	0	2
CO4	3	3	3	2	3	0	0	1	0	0	2
Course Content										No of Lectures	
Unit I: Database Design and ER Modeling: Entity-Relationship (ER) Model, ER Diagram Design, Mapping ER Model to Relational Schema, Database Design for Real-world Applications. Table Creation and Constraints: SQL Table Creation, PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL, CHECK and DEFAULT Constraints, DDL and DML Operations. DDL Commands: CREATE, ALTER, DROP, TRUNCATE. DML Commands: INSERT, UPDATE, DELETE, Data Retrieval using SELECT.										[6]	
Unit II: SQL Queries and Functions: Aggregate Functions (COUNT, SUM, AVG, MAX, MIN), String Functions, Date Functions, Mathematical Functions. Joins and Set Operations: INNER JOIN, LEFT JOIN, RIGHT JOIN, SELF JOIN, UNION, INTERSECT MINUS/EXCEPT, Grouping and Subqueries: ORDER BY, GROUP BY, HAVING Clause, Nested Subqueries, EXISTS, ANY, ALL Operators.										[6]	
Unit III: Views and Integrity Constraints: Creating, Modifying and Dropping Views, Entity Integrity, Referential Integrity, Data Consistency and Validation. PL/SQL Programming: Variables and Constants, Conditional Statements (IF-ELSE, CASE), Iterative Statements (FOR, WHILE), Exception Handling.										[6]	

Approved by BoS of USAR : 16/6/2026

Approved by AC sub-committee : 20/07/2026

Applicable for Batch admitted in 2025 and Lateral Entry of AS 2026-27 admitted in Batch 2025

Unit IV: Stored Procedures, Functions and Cursors: Creation and Execution of Procedures, User-defined Functions, Implicit and Explicit Cursors, Record Processing. Triggers: BEFORE Triggers, AFTER Triggers, INSERT, UPDATE and DELETE Triggers. Development of a Mini Database Application integrating: Tables, Queries, Views, Procedures, Triggers.	[6]
Text Books [T1] Sadalage, P . J., & Fowler, M. (2013). NoSQL distilled: a brief guide to the emerging world of polyglot persistence (1st ed.). Pearson Education. [T2] Silberschatz, A., Korth, H. F., & Sudarshan, S. (2002). Database system concepts (4th ed.). New York: McGraw-Hill. [T3] Elmasri, R., Navathe, S. B., Elmasri, R., & Navathe, S. B. (2000). Fundamentals of Database Systems (3rd ed.). Addison-Wesley.	
Reference Books [R1] Date, C. J. (2004). An Introduction to Database Systems. (8-th ed). [R2] Ullman, J. D. (1983). Principles of database systems. (1st ed) Galgotia publications. [R3] Bipin C. Desai. (1990). An Introduction to Database Systems (1st ed). West Publishing Co.	

Paper code: ARI261							L		P		Credit
Paper Title: OOPS with JAVA lab							0		2		1
Marking Scheme: Internal Evaluation: As per university examination norms from time to time. End Term Practical Examination: As per university examination norms from time to time.											
Course Outcomes [Bloom’s Knowledge Level (KL)]: CO1: Understand how to implement Object Oriented concepts through Java. [K2] CO2: Understand Inheritance and apply the basics of Java 8 Constructs in inheritance. [K2] CO3: Identify and apply the Java Inheritance to program Java applications. [K2, K3] CO4: Identify and apply the Java Exception handling to program Java applications. [K2, K3]											
CO/PO	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11
CO1	2	3	2	3	3	-	-	-	-	-	3
CO2	2	3	3	3	3	-	-	-	-	-	3
CO3	2	3	3	3	3	-	-	-	-	-	3
CO4	2	3	3	3	3	-	-	-	-	-	3
Unit-I Java Basics: Difference between Java and C++, Java Tokens. Basics of Java programming: Data types, Literals, Variables, Scope and lifetime of variables, Operators. Control Structures including selection, Looping, Arrays.											[6]
Unit-II Introducing Classes: Creating a Class: properties, methods and constructors. Object Access modifiers, this keyword, final keyword.											[6]
Unit-III Inheritance: Types, Super keyword, method overriding, abstract class. Interfaces: Creation and implementing an interface, difference between abstract class and interface.											[6]
Unit-IV Exception Handling: Exception Class, built-in checked and unchecked exceptions, user defined exceptions, use of try, catch, throw, throws, finally.											[6]
Text Books: [T1] Schildt, H. (2021). Java: The Complete Reference (12th ed.). McGraw Hill. [T2] Sierra, K., Bates, B., & Gee, T. (2022). Head First Java: A Brain-Friendly Guide (3rd ed.). O'Reilly Media.											
Reference Books: [R1] Balaguruswamy, E. (2019). Programming with Java: A Primer (7th ed.). McGraw Hill Education. [R2] Horstmann, C. S. (2000). Computing Concepts with Java 2 Essentials (2nd ed.). John Wiley & Sons. [R3] Decker, R., & Hirshfield, S. (2001). Programming Java: An Introduction to Programming Using Java (2nd ed.). Vikas Publishing House.											

DETAILED SYLLABUS FOR 4th SEMESTER

Paper Code: ARI202								L	P	Credit	
Paper Title: Internet of Things Architecture and Protocol								3	0	3	
Marking Scheme: Teachers Continuous Evaluation: As per university examination norms from time to time. End Term Theory Examination: As per university examination norms from time to time.											
INSTRUCTIONS TO PAPER SETTERS: Maximum Marks : As per University norms											
➤ There should be 9 questions in the end term examination question paper ➤ Question No. 1 should be compulsory and cover the entire syllabus. This question should have objective or short answer type questions. ➤ Apart from Question No. 1, the rest of the paper shall consist of four units As per the syllabus. Every unit should have two questions. However, students may be asked to attempt only 1 question from each unit. ➤ The questions are to be framed keeping in view the learning outcomes of course/paper. The standard/ level of the questions to be asked should be at the level of the prescribed textbooks. ➤ The requirement of (scientific) calculators/ log-tables/ data-tables may be specified if required											
Course Outcomes [Bloom’s Knowledge Level (KL)]: CO1: To understand the and ecosystem and hardware peripherals of the Industrial Internet of Things (IIoT). [K1, K2] CO2: To understand and implement industrial communication protocols at network and application layer. [K3] CO3: To analyze core architecture, reference models, cloud platforms and develop basic predictive maintenance frameworks. [K4] CO4: To evaluate cybersecurity risks and apply functional safety standards (IEC 61508/62443) to automation networks. [K4, K5]											
CO/PO	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11
CO1	3	3	3	3	1	-	-	-	-	1	2
CO2	3	3	3	3	1	2	-	-	-	1	2
CO3	3	3	3	3	1	2	-	-	1	2	3
CO4	3	3	3	3	1	2	-	-	2	2	3
Course Content										No of Lectures	
Unit-I IoT Fundamentals and System Programming: IoT Ecosystem and Core components: smart sensors, actuators, edge gateways, and industrial cloud infrastructure, Evolution from M2M, Edge computing vs. cloud computing paradigms, data ingestion pipelines, and processing topologies, IoT and M2M value chain. Introduction to programming using Embedded C/C++ and hardware system design utilizing Arduino/Raspberry Pi, and ESP32/ESP8266 platforms with emphasis on sensor interfacing, output regulation, and pin-level input processing.										[8]	
Unit-II IoT Communication Protocols: Modern Industrial Ethernet standards: Profinet, EtherCAT,										[8]	

Approved by BoS of USAR : 16/6/2026

Approved by AC sub-committee : 20/07/2026

Applicable for Batch admitted in 2025 and Lateral Entry of AS 2026-27 admitted in Batch 2025

Modbus TCP. Digital sensor communication, IEEE802.15.4–BACNet Protocol, KNX, ZigBee protocol architecture. IoT network layer protocol: 6LoWPAN (IPv6 over Low-Power Wireless Personal Area Networks), Threads, RPL (Routing Protocol for Low-Power and Lossy Networks), Cellular IoT Extensions (NB-IoT & LTE-M). IoT application layer protocols: HTTP, MQTT (Message Queuing Telemetry Transport), CoAP (Constrained Application Protocol), AMQP.	
Unit-III IoT Architecture and Cloud Platforms: IoT Open-source architecture (OIC) & Design principles, Interoperability and standardization using OPC UA. Industrial Internet Reference Architecture (IIRA), RAMI 4.0. IoT Prototyping Frameworks: Android Things, ThingSpeak: cloud-based data storage and analytics platform, IoTivity: An Open source IoT stack Overview, Resource modelling with Supported Operational Methods (CRUDN) and Abstraction, Node-RED: Visual Architecture (Flow-Based Programming). IoT cloud platforms: AWS IoT Core, Microsoft Azure IoT and Blynk IoT Platform.	[8]
Unit-IV Web of Things and Functional Safety: Architecture Standardization and Platform Middleware for WoT, Unified Multitier WoT Architecture, REST (Representational State Transfer), WoT Portals and Business Intelligence. Cybersecurity vulnerabilities in IT-OT convergence, IoT Security Tomography, threat landscapes, common attack vectors, and defense-in-depth strategies. Industrial security standards: IEC 62443 overview. Functional safety in automated environments: introduction to IEC 61508 and IEC 61511. Safety Integrity Levels (SIL) assessment, risk matrix, fail-safe architectures, and emergency shutdown system integration.	[8]
Text Books [T1] Gilchrist, A. (2016). Industry 4.0: The Industrial Internet of Things (1st ed.). Apress. [T2] Misra, S., Roy, C., & Mukherjee, A. (2021). Introduction to Industrial Internet of Things and Industry 4.0 (1st ed.). CRC Press. [T3] Cirani, S., Ferrari, G., Picone, M., & Veltri, L. (2019). Internet of Things: Architectures, Protocols, and Standards (1st ed.). John Wiley & Sons.	
Reference Books [R1] Lea, P. (2018). IoT Architecture and Design Handbook (2nd ed.). Packt Publishing.	

Paper code: ARI204								L	P	Credit	
Paper Title: PLC-SCADA								3	0	3	
Marking Scheme: Teachers Continuous Evaluation: As per university examination norms from time to time. End Term Theory Examination: As per university examination norms from time to time.											
INSTRUCTIONS TO PAPER SETTERS: Maximum Marks: As per University norms											
➤ There should be 9 questions in the end term examination question paper ➤ Question No. 1 should be compulsory and cover the entire syllabus. This question should have objective or short answer type questions. ➤ Apart from Question No. 1, the rest of the paper shall consist of four units As per the syllabus. Every unit should have two questions. However, students may be asked to attempt only 1 question from each unit. ➤ The questions are to be framed keeping in view the learning outcomes of course/paper. The standard/ level of the questions to be asked should be at the level of the prescribed textbooks. ➤ The requirement of (scientific) calculators/ log-tables/ data-tables may be specified if required.											
Course Outcomes [Bloom’s Knowledge Level (KL)]: CO1: Ability of students to understand the fundamental concepts of PLC and controlled hardware. [K1, K2] CO2: Ability of students to understand the basic programming concepts of PLC. [K1, K2, K3] CO3: Ability of students to understand the basics concepts of SCADA system and utilized communication protocols for analyze the system [K1, K2, K3] CO4: Ability of students to understand the basics concepts SCADA programming and PLC interfacing and find industrial applications [K1, K2, K4].											
CO / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	3	3	1	-	-	1	-	3
CO2	3	3	3	3	3	1	-	-	1	-	3
CO3	3	3	3	3	3	1	-	-	1	-	3
CO4	3	3	3	3	3	1	-	-	1	-	3
Course Content											No of Lectures
Unit-I PLC Fundamental and Hardware: Evolution of PLC; PLC based automation; Relay logic and comparison; Architecture of PLC; Types of PLC; Features of PLC; Power supplies and isolators; Scan cycle; I/O Modules and classification of I/O field devices; Bit instructions; Sinking/ Sourcing wiring concept; PLC interfacing; Introduction to RSLinx Software; Introduction to RSLogix 500 Software; PLC Networking.											[8]

Unit-II PLC Programming: Ladder logic; Function block diagram; Structured; Ladder logic essentials: Contacts (NO/NC) and basic logic gates; Advanced instructions: Registers, Timers: On-delay, Off-delay, retentive timers; Counters: up-counters, down-counters (CTD), reset instructions; Data handling: Math instructions, data move, comparison blocks; Process controllers and loop tuning using PLC.	[8]
Unit III SCADA System and Communication Protocols: Introduction to SCADA system; SCADA architecture: Master terminal units, remote terminal units, communication servers; Human machine interface; Alarm management: Real-time and historical; Real-time and historical data logging for process analysis; Database integration; Industrial Networking: Physical media and network topologies; Communication Protocols: Deep dive into RS-232, Modbus (RTU/TCP) and Ethernet/IP.	[8]
Unit IV SCADA: Programming and Applications: SCADA programming; Interfacing of SCADA and PLC; Remote operation monitoring using SCADA; SCADA interfaced PLCs to flow process station; SCADA interfaced PLCs to pH control system process station; SCADA interfaced PLCs to split control system process station; SCADA interfaced PLCs to temperature process station.	[8]
Text Books [TI] AlMadhoun, A. S. (2025). PLC SCADA for Beginners: Understanding and Implementing Industrial Automation Systems (1st ed.). Springer Nature.	
Reference Books [R1] Akande, O. (2023). PLC Industrial Automation from Scratch: A Hands-On Guide (1st ed.). Packt Publishing.	

Paper code: ARI206								L	P	Credit	
Paper Title: Signals and Systems								3	0	3	
Marking Scheme: Teachers Continuous Evaluation: As per university examination norms from time to time. End Term Theory Examination: As per university examination norms from time to time.											
INSTRUCTIONS TO PAPER SETTERS: Maximum Marks: As per University norms											
➤ There should be 9 questions in the end term examination question paper ➤ Question No. 1 should be compulsory and cover the entire syllabus. This question should have objective or short answer type questions. ➤ Apart from Question No. 1, the rest of the paper shall consist of four units As per the syllabus. Every unit should have two questions. However, students may be asked to attempt only 1 question from each unit. ➤ The questions are to be framed keeping in view the learning outcomes of course/paper. The standard/ level of the questions to be asked should be at the level of the prescribed textbooks. ➤ The requirement of (scientific) calculators/ log-tables/ data-tables may be specified if required											
Course Outcomes [Bloom’s Knowledge Level (KL)]: CO1: Ability of students to understand and represent signals and perform basic operations on signals. [K1, K2] CO2: Ability of students to apply and analyze different real-life periodic and aperiodic signals to system. [K1, K2, K3] CO3: Ability of students to provide valuable insights of complex systems/signals analyzed through different techniques learnt. [K1, K2, K3] CO4: Ability of students to understand and analyze different types of signals and systems and transformation techniques for future courses in Signal Processing, Image processing and so on. [K1, K3, K4]											
CO/PO	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11
CO1	3	3	3	3	2	-	-	-	-	-	2
CO2	3	3	3	3	2	-	-	2	1	-	2
CO3	3	3	3	2	3	-	-	2	1	-	1
CO4	3	3	3	2	3	-	-	1	1	-	1
Course Content											No of Lectures
Unit I Signals – Signals and their representation: Continuous-Time and Discrete-Time Signals- Unit Impulse, Unit Step, Ramp, Exponential & Sinusoidal Signals. Periodic & aperiodic signals, Deterministic and random signals, Energy and Power signals. Continuous-Time and Discrete-Time Systems. Classification, Static & dynamic, Linear and non-linear, Causal and non-causal, Time variant and invariant.											[8]
Unit II Analysis of Continuous & Discrete time signals: Fourier series Representation of Continuous-											[8]

Time Periodic Signals, Properties, Continuous-Time Fourier Transform (CTFT), The Fourier Transform for Periodic Signals, Properties of the CTFT, Duality, Sinc and signum function, Sampling Theorem, Aliasing, Discrete Time Fourier series Properties, Discrete-Time Fourier Transform (DTFT). Properties of the DTFT. Parseval's Theorem, Central ordinate theorem.	
Unit III Convolution and Correlation of Signals: Concept of convolution in time domain, Cross correlation and auto correlation of functions, properties of correlation function, Energy density spectrum, Parseval's theorem, Power density spectrum, Relation between convolution and correlation.	[8]
Unit IV Discrete Time Signals & Systems: Discrete time signals, representation, operations on sequences, discrete time systems and classification, LTI systems, Linear Convolution, Difference equations. Z-Transforms: ROC, properties of Z-Transforms Inverse Z-Transforms, Causality and stability.	[8]
Text Books: [T1] Oppenheim, A. V., Willsky, A. S., & Nawab, S. H. (2015). Signals and Systems (2nd ed., Pearson India ed.). Pearson India. [T2] Roberts, M. J., & Sharma, G. (2017). Fundamentals of Signals and Systems (2nd ed.). McGraw Hill Education.	
Reference Books: [R1] Haykin, S., & Van Veen, B. (2007). Signals and Systems (2nd ed.). John Wiley & Sons. [R2] Lathi, B. P. (2005). Linear Systems and Signals (2nd ed.). Oxford University Press. [R3] Ziemer, R. E., Tranter, W. H., & Fannin, D. R. (2014). Signals and Systems: Continuous and Discrete (4th ed.). Pearson.	

Paper code: ARI208							L	P	Credit		
Paper Title: Artificial Intelligence and Machine Learning for IIOT							3	0	3		
Marking Scheme:											
Teachers Continuous Evaluation: As per university examination norms from time to time.											
End Term Theory Examination: As per university examination norms from time to time.											
INSTRUCTIONS TO PAPER SETTERS: Maximum Marks: As per University norms											
➤ There should be 9 questions in the end term examination question paper. ➤ Question No. 1 should be compulsory and cover the entire syllabus. This question should have objective or short answer type questions. ➤ Apart from Question No. 1, the rest of the paper shall consist of four units As per the syllabus. Every unit should have two questions. However, students may be asked to attempt only 1 question from each unit. ➤ The questions are to be framed keeping in view the learning outcomes of course/paper. The standard/ level of the questions to be asked should be at the level of the prescribed textbooks. ➤ The requirement of (scientific) calculators/ log-tables/ data-tables may be specified if required.											
Course Outcomes [Bloom’s Knowledge Level (KL)]:											
CO1: To Understand Agent development. [K1, K2]											
CO2: To Gain Knowledge in Multi agent and Intelligent agents. [K3, K4]											
CO3: To Understand, apply and analyse regression and ANN. [K2, K4]											
CO4: To apply the concepts of ML algorithms to various applications. [K3, K4]											
CO/PO	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11
CO1	3	3	3	2	3	1	-	2	-	1	1
CO2	3	2	3	2	3	1	-	2	-	1	1
CO3	3	2	3	3	3	1	-	2	-	1	1
CO4	3	2	3	3	3	1	-	2	-	1	1
Course Content											No of Lectures
Unit I: Intelligent Agents Introduction, AI problems, Agents & environments, Concept of rationality, Structure of agents, Simple reflex agents, Model-based reflex agents, Goal based agents, Utility based agents, learning agents, learning processes (supervised, unsupervised, semi- supervised/reinforcement learning), Learning tasks, Distance metrics.											[8]
Unit II: Searching Searching for solutions, uniformed search strategies – Breadth first search, depth first Search. Search with partial information (Heuristic search) Hill climbing, A*, AO* Algorithms, Problem reduction, Game Playing-Adversial search, Games, mini-max algorithm, optimal decisions in multiplayer games, Problem in Game playing, Alpha-Beta pruning, Evaluation functions.											[8]

Unit III: Introduction to Artificial Neural Networks Linear separability, Non-linear problems, Artificial Neuron, perceptron, Adaptive linear neurons, perceptron convergence theorem, gradient descent algorithm, support vector machine, radial basis functions.	[8]
Unit IV: Introduction to Machine Learning Machine learning tasks, linear regression, regularized regression, logistic regression, clustering techniques. K-means clustering, Expectation maximization algorithm; Principal Component Analysis and Linear Discriminant Analysis.	[8]
Text Books: [T1] Russell, S., & Norvig, P. (2021). Artificial Intelligence: A Modern Approach (4th Global ed.). Pearson. [T2] Haykin, S. (2009). Neural Networks and Learning Machines (3rd ed.). Pearson. [T3] Zurada, J. M. (1997). Introduction to Artificial Neural Systems (Indian ed.). Jaico Publishing House.	
Reference Books: [R1] Cichocki, A., & Unbehauen, R. (1993). Neural Networks for Optimization and Signal Processing (1st ed.). John Wiley & Sons. [R2] Mitchell, T. M. (1997). Machine Learning (1st ed.). McGraw-Hill.	

Paper code: ARI210								L	P	Credit	
Paper Title: Control Systems								3	0	3	
Marking Scheme: Teachers Continuous Evaluation: As per university examination norms from time to time. End Term Theory Examination: As per university examination norms from time to time.											
INSTRUCTIONS TO PAPER SETTERS: Maximum Marks: As per University norms											
➤ There should be 9 questions in the end term examination question paper. ➤ Question No. 1 should be compulsory and cover the entire syllabus. This question should have objective or short answer type questions. ➤ Apart from Question No. 1, the rest of the paper shall consist of four units As per the syllabus. Every unit should have two questions. However, students may be asked to attempt only 1 question from each unit. ➤ The questions are to be framed keeping in view the learning outcomes of course/paper. The standard/ level of the questions to be asked should be at the level of the prescribed textbooks. ➤ The requirement of (scientific) calculators/ log-tables/ data-tables may be specified if required.											
Course Outcomes [Bloom’s Knowledge Level (KL)]: CO1: Ability of students to utilize concepts of control system components and mathematical modeling of electrical system, mechanical system, etc. [K1, K2, K3] CO2: Ability of students to identify and implement the concept of time response and frequency response of the system. [K1, K3] CO3: Ability of students to utilize understanding of different plots such as Bode plot, Nyquist plot, Root locus method and Polar plot and implement them for robot applications. [K2, K3, K4] CO4: Ability of students to practically implement knowledge on joint space and task space control scheme in robots. [K3, K4]											
CO/PO	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11
CO1	3	3	3	3	2	-	-	1	1	-	3
CO2	3	3	3	3	2	-	-	1	1	-	3
CO3	3	3	3	3	-	-	-	1	1	-	3
CO4	3	3	3	3	-	-	-	1	1	-	3
Course Content											No of Lectures
Unit-I Introduction to Control System: Concept of open loop and closed loop systems, Effects of feedback, Transfer function of electrical, mechanical (translational and rotational) System. Force Voltage and Force Current analogies. Block diagram reduction technique and signal flow graph, Mason’s rule, Signal flow graph of electrical network.											[8]
Unit-II Time Domain Analysis: Time domain analysis of standard first and second order closed loop system. Concept of un-damped natural frequency, damping, overshoot, rise time and settling time. Dependence of time domain performance parameters on natural frequency and damping											[8]

ratio. Step and Impulse response of first and second order systems, Stability by pole location. Routh-Hurwitz criteria and applications. Error Analysis: Steady state errors in control systems due to step, ramp and parabolic inputs. Concepts of system types and error constants.	
Unit-III Root locus Techniques: Definition and properties of root locus, rules for plotting root locus, stability analysis using root locus. Frequency Domain Analysis: Bode plot, Polar plots, stability in frequency domain, Nyquist plots, Nyquist stability criterion, Gain margin and phase margin via Nyquist diagram.	[8]
Unit-IV State Variable Analysis: Introduction to state variable, General state space representation, Introduction to controllability, controllability matrix, observability, observability matrix. Introduction to Compensator: Basic concept of compensator design, requirement, lead, lag, and lead-lag compensator, proportional, derivative, integral, P-D, P-I and P-I-D controller.	[8]
Text Books [T1] Nise, N. S. (2024). Control Systems Engineering (8th ed.). John Wiley & Sons. [T2] Kuo, B. C., & Golnaraghi, F. (2017). Automatic Control Systems (10th ed.). McGraw Hill Education. [T3] Wilkie, J., Johnson, M., & Katebi, R. (2002). Control Engineering: An Introductory Course. Palgrave Macmillan (2nd ed.). [T4] Ogata, K. (2020). Modern Control Engineering (6th ed.). Pearson.	
Reference Books [R1] Dorf, R. C., & Bishop, R. H. (2021). Modern Control Systems (14th ed.). Pearson. [R2] D'Azzo, J. J., Houpis, C. H., Sheldon, S. N., & Dekkar, M. (2003). Linear Control System Analysis and Design with MATLAB (5th ed.). CRC Press. [R3] Manesis, S., & Nikolakopoulos, G. (2018). Introduction to Industrial Automation (1st ed.). CRC Press.	

DETAILED SYLLABUS FOR 5th SEMESTER

Paper code: ARI301								L	P	Credit	
Paper Title: Electronic Design Automation for VLSI								3	0	3	
Marking Scheme: Teachers Continuous Evaluation: As per university examination norms from time to time. End Term Theory Examination: As per university examination norms from time to time.											
INSTRUCTIONS TO PAPER SETTERS: Maximum Marks: As per University norms											
➤ There should be 9 questions in the end term examination question paper. ➤ Question No. 1 should be compulsory and cover the entire syllabus. This question should have objective or short answer type questions. ➤ Apart from Question No. 1, the rest of the paper shall consist of four units As per the syllabus. Every unit should have two questions. However, students may be asked to attempt only 1 question from each unit. ➤ The questions are to be framed keeping in view the learning outcomes of course/paper. The standard/ level of the questions to be asked should be at the level of the prescribed textbooks. ➤ The requirement of (scientific) calculators/ log-tables/ data-tables may be specified if required											
Course Outcomes [Bloom’s Knowledge Level (KL)]: CO1: Students will understand and define various aspects of VLSI physical design and automation. [K1, K2] CO2: The ability of students to understand the VLSI fabrication process. [K1, K2] CO3: Illustrating the EDA simulator for circuit design and circuit simulation process. [K3, K4] CO4: Understand, apply and analyze the layout designing of various VLSI circuits and devices. [K2, K3, K4]											
CO/PO	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11
CO1	3	3	2	3	3	-	-	-	1	-	3
CO2	3	3	2	3	3	-	-	-	1	-	3
CO3	3	3	2	3	3	-	-	-	1	-	3
CO4	3	3	3	3	3	-	-	-	1	-	3
Course Content										No of Lectures	
Unit-I Physical Design Automation: Basics of VLSI automation, Design cycle: system specifications, architectural, behavioral, logic, circuit, & physical design, fabrication, packaging, testing and debugging, new trends in VLSI design cycle- Increasing: interconnect delay, interconnect area, number of metal layers, planning requirements. Logic synthesis, high-level synthesis, Physical design cycle: Partitioning, Floor-planning and Placement, Routing, Extraction and Verification, New trends in physical design cycle: chip level signal planning, OTC routing, Design Styles: Full custom, standard cell, Gate array, Field programmable gate array, sea of gates, Comparison of Different Design Styles.										[8]	
Unit-II Fabricaiton of CMOS ICs: Fabrication materials, MOS architecture, Fabrication of integrated										[8]	

circuits, material growth and oxidation: silicon dioxide, silicon nitride, Polycrystalline silicon, metals, doped silicon layers: diffusion & ion implantation, chemical mechanical polishing, Lithography: clean room, nMOS, pMOS fabrication steps, CMOS process flow, field oxide, shallow trench isolation.	
Unit-III Circuit simulator: Simulator basics and type of simulators, historical perspective, circuit simulations: DC analysis: sweeping a source, the .dc statement, printing output, plotting output, graphics output, subcircuits, Ac analysis: specifying input source, Plotting bode plot, plotting group delay, input impedance, plotting output impedance, Noise analysis: the .noise statement, print and plot output, signal to noise, inserting noise source, Transient analysis: Simulating time, specifying input source, the .trans statement, graphic output and calculation, setting initial conditions, transient solution for static problems.	[8]
Unit-IV Layout Simulation: MOSFET Scaling and short channel effects, Layout design rules: micron & lambda rules: size rules, separation rules, overlap rules, Layouts of basic devices: nMOS, pMOS, Basic gate design: CMOS Inverter, NAND, NOR, Memory cells: 6T SRAM, DRAM. Basics of EDA tools: Layout and basics of simulators: Layout editor, Extraction, Design rule check, Layout versus schematic, Electrical Rule check.	[8]
Textbooks: [T1] Sherwani, N. A. (2013). Algorithms for VLSI Physical Design Automation (3rd ed.). Springer. (Reprint) [T2] Uyemura, J. P. (2002). Introduction to VLSI Circuits and Systems (1st ed.). John Wiley & Sons. [T3] Rashid, M. H. (2015). SPICE for Circuits and Electronics Using PSpice (3rd ed.). Pearson.	
Reference Books: [R1] Sze, S. M. (1988). VLSI Technology (2nd ed.). McGraw-Hill. [R2] Kundert, K. S. (2013). The Designer's Guide to SPICE and Spectre. Springer. (Softcover reprint of the 1995 edition.)	

Paper code: ARI303								L	P	Credit	
Paper Title: Analog and Digital Communication Systems								3	0	3	
Marking Scheme: Teachers Continuous Evaluation: As per university examination norms from time to time. End Term Theory Examination: As per university examination norms from time to time.											
INSTRUCTIONS TO PAPER SETTERS: Maximum Marks: As per University norms											
➤ There should be 9 questions in the end term examination question paper. ➤ Question No. 1 should be compulsory and cover the entire syllabus. This question should have objective or short answer type questions. ➤ Apart from Question No. 1, the rest of the paper shall consist of four units As per the syllabus. Every unit should have two questions. However, students may be asked to attempt only 1 question from each unit. ➤ The questions are to be framed keeping in view the learning outcomes of course/paper. The standard/ level of the questions to be asked should be at the level of the prescribed textbooks. ➤ The requirement of (scientific) calculators/ log-tables/ data-tables may be specified if required											
Course Outcomes [Bloom’s Knowledge Level (KL)]: CO1: Understand the fundamental principles of continuous-wave amplitude modulation and analyze noise performance in communication systems. [K2, K4] CO2: Analyze the generation, transmission bandwidth, and detection methods of angle modulation systems (FM and PM). [K2, K3, K4] CO3: Apply the sampling theorem to evaluate and compare various analog and digital pulse modulation techniques. [K3, K4] CO4: Evaluate the generation, detection, and error probability performance of digital modulation schemes over AWGN channels. [K4, K5]											
CO/PO	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11
CO1	3	2	-	-	1	-	-	1	-	-	2
CO2	3	2	-	-	1	-	-	1	-	-	2
CO3	3	3	3	-	-	-	-	1	1	2	2
CO4	3	2	-	3	-	-	-	1	-	-	2
Course Content											No of Lectures
Unit-I Amplitude Modulation (AM): Introduction to communication systems, Need for modulation, Time and Frequency domain description of AM, Generation of AM waves (Switching modulator), Detection (Envelope detector). Suppressed Carrier Techniques: DSB-SC, SSB-SC, and VSB (Vestigial Side Band) modulation—their generation and coherent detection. Noise in Communication: Noise sources, Signal-to-Noise Ratio (SNR), Noise figure and Noise temperature.											[8]

Unit-II Angle Modulation: Concepts of Frequency Modulation (FM) and Phase Modulation (PM), Single-tone FM, Narrowband and Wideband FM, Transmission bandwidth (Carson's Rule). Generation & Detection: Armstrong method for FM generation, FM Demodulation using Phase Locked Loops (PLL) and Foster-Seeley discriminators.	[8]
Unit-III Sampling Process: Sampling theorem for low-pass and band-pass signals, Aliasing, Aperture effect. Pulse Analog Modulation: PAM (Pulse Amplitude Modulation), PWM (Pulse Width Modulation), and PPM (Pulse Position Modulation)—their generation and detection. Pulse Digital Modulation: Elements of Pulse Code Modulation (PCM), Quantization (Uniform & Non-uniform), Differential PCM (DPCM). Delta Modulation: Delta Modulation (DM), Adaptive Delta Modulation (ADM), and limitations.	[8]
Unit-IV Digital Modulation Schemes: Generation and Detection of Amplitude Shift Keying (ASK), Frequency Shift Keying (FSK), and Phase Shift Keying (PSK). Performance Analysis: Probability of bit error calculations for BPSK, BFSK, and QPSK in the presence of AWGN (Additive White Gaussian Noise).	[8]
Text Books: [T1] Haykin, S., Moher, M., & Koilpillai, R. D. (2022). Communication Systems (5th ed., Indian adaptation). Wiley India. [T2] Lathi, B. P., & Ding, Z. (2019). Modern Digital and Analog Communication Systems (5th ed.). Oxford University Press. [T3] Taub, H., Schilling, D. L., & Saha, G. (2019). Principles of Communication Systems (3rd ed.). McGraw Hill Education.	
Reference Books: [R1] Davis, K., & Davis, M. (2019). Electronic Communication Systems (6th ed., Special Indian Edition). McGraw Hill Education. [R2] Frenzel, L. E. (2024). Principles of Electronic Communication Systems (5th ed.). McGraw Hill. [R3] Roddy, D., & Coolen, J. (2008). Electronic Communications (4th ed., Indian Subcontinent adaptation). Pearson Education India	

Paper code: ARI305								L	P	Credit	
Paper Title: Wireless Sensor Networks								3	0	3	
Marking Scheme: Teachers Continuous Evaluation: As per university examination norms from time to time. End Term Theory Examination: As per university examination norms from time to time.											
INSTRUCTIONS TO PAPER SETTERS: Maximum Marks: As per University norms											
➤ There should be 9 questions in the end term examination question paper. ➤ Question No. 1 should be compulsory and cover the entire syllabus. This question should have objective or short answer type questions. ➤ Apart from Question No. 1, the rest of the paper shall consist of four units As per the syllabus. Every unit should have two questions. However, students may be asked to attempt only 1 question from each unit. ➤ The questions are to be framed keeping in view the learning outcomes of course/paper. The standard/ level of the questions to be asked should be at the level of the prescribed textbooks. ➤ The requirement of (scientific) calculators/ log-tables/ data-tables may be specified if required											
Course Outcomes [Bloom’s Knowledge Level (KL)]: CO1: Ability of students to understand the fundamentals of wireless communication technologies and cellular system fundamentals. [K1, K2] CO2: Ability of students to understand the fundamentals of wireless sensor networks and its application to critical real time scenarios. [K1, K2, K3] CO3: Ability of students to identify and formulate various protocols at various layers and its differences with traditional protocols for WSN. [K1, K2, K3] CO4: Ability of students to implement the routing protocols and challenges involved in managing a sensor networks. [K3, K4, K5]											
CO/PO	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11
CO1	3	3	3	2	2	-	-	-	1	-	2
CO2	3	3	2	3	3	-	-	1	1	-	2
CO3	3	3	2	3	3	-	-	1	1	-	3
CO4	3	3	2	3	3	-	-	1	1	-	3
Course Content											No of Lectures
Unit I Fundamentals of Wireless Communication Technology: Electromagnetic spectrum radio propagation, Wireless Communication Systems: Paging Systems, Cordless Telephone Systems, Cellular Telephone Systems, Modern Wireless Communication Systems: Overview of 5G and 6G wireless technology, Cellular system design fundamentals: Frequency Reuse, Channel Assignment Strategies, Handoff Strategies, Interference and System Capacity, Trunking and Grade of Service, Improving Coverage & Capacity in Cellular Systems.											[8]
Unit II Adhoc/Sensor Networks: Introduction to adhoc/sensor networks: Key definitions of adhoc/											[8]

Approved by BoS of USAR : 16/6/2026

Approved by AC sub-committee : 20/07/2026

Applicable for Batch admitted in 2025 and Lateral Entry of AS 2026-27 admitted in Batch 2025

sensor networks, unique constraints and challenges, advantages of ad-hoc/sensor network, driving applications, issues in adhoc wireless networks, issues in design of sensor network, sensor network architecture, data dissemination and gathering.	
Unit III MAC Protocols: Issues in designing MAC protocols for adhoc wireless networks, design goals, classification of MAC protocols, MAC protocols for sensor network, location discovery, quality of a Sensor Network, other issues, S-MAC, IEEE 802.15.4.	[8]
Unit IV Routing Protocols and QoS: Issues in designing a routing protocol, classification of routing protocols, table-driven, on-demand, hybrid, flooding, hierarchical, and power aware routing protocols, Issues and Challenges in providing QoS, QoS frameworks, need for energy management, and system power management schemes, Case study of Real time application.	[8]
Text Books: [T1] Murthy, C. S. R., & Manoj, B. S. (2004). Ad Hoc Wireless Networks: Architectures and Protocols (1st ed.). Pearson Education. [T2] Rappaport, T. S. (2024). Wireless Communications: Principles and Practice (2nd ed.). Cambridge University Press. [T3] Zhao, F., & Guibas, L. J. (2004). Wireless Sensor Networks: An Information Processing Approach (1st ed.). Elsevier.	
Reference Books: [R1] Schiller, J. (2008). Mobile Communications (2nd ed.). Pearson Education. [R2] Stallings, W. (2005). Wireless Communications and Networks (2nd ed.). Pearson Education. [R3] Goldsmith, A. (2023). Wireless Communications (2nd ed.). Cambridge University Press.	

Paper code: ARI371T								L	P	Credit	
Paper Title: Deep Learning								3	0	3	
Marking Scheme: Teachers Continuous Evaluation: As per university examination norms from time to time. End Term Theory Examination: As per university examination norms from time to time.											
INSTRUCTIONS TO PAPER SETTERS: Maximum Marks: As per University norms											
➤ There should be 9 questions in the end term examination question paper. ➤ Question No. 1 should be compulsory and cover the entire syllabus. This question should have objective or short answer type questions. ➤ Apart from Question No. 1, the rest of the paper shall consist of four units As per the syllabus. Every unit should have two questions. However, students may be asked to attempt only 1 question from each unit. ➤ The questions are to be framed keeping in view the learning outcomes of course/paper. The standard/ level of the questions to be asked should be at the level of the prescribed textbooks. ➤ The requirement of (scientific) calculators/ log-tables/ data-tables may be specified if required											
Course Outcomes [Bloom’s Knowledge Level (KL)]: CO1: Explain the fundamentals of neural networks and deep learning architectures. [K2] CO2: Apply optimization and regularization techniques for training deep neural networks. [K3] CO3: Analyze convolutional neural network architectures for computer vision applications. [K4] CO4: Apply deep learning models for sequence processing and real-world applications. [K3]											
CO/PO	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11
CO1	3	2	-	-	1	-	-	-	-	-	1
CO2	3	3	2	2	3	-	-	-	-	-	1
CO3	3	3	3	2	3	-	-	-	-	-	1
CO4	3	3	2	2	3	-	-	-	-	-	1
Course Content											No of Lectures
Unit-I Foundations of Deep Learning: Biological Neuron vs. Artificial Neuron, Introduction to Artificial Neural Networks, Perceptron and Multi-Layer Perceptron (MLP), Activation Functions: Sigmoid, Tanh, ReLU, Softmax, Feedforward Neural Networks, Loss Functions and Cost Optimization, Gradient Descent and Variants, Backpropagation Algorithm, Challenges in Deep Neural Networks: Vanishing and Exploding Gradients.											[8]
Unit-II Deep Neural Networks and Optimization: Deep Feedforward Networks, Weight Initialization Techniques, Regularization Methods: L1, L2, Dropout, Batch Normalization, Hyperparameter Tuning, Optimization Algorithms: SGD, Momentum, RMSProp, Adam, Model Evaluation and Performance Metrics.											[8]

Unit-III: Convolutional Neural Networks (CNNs): Introduction to CNNs, Convolution and Pooling Operations, CNN Architecture and Layers, Popular CNN Architectures: LeNet, AlexNet, VGG, Transfer Learning Image Classification Applications, Object Detection.	[8]
Unit-IV Recurrent Neural Networks and Applications: Sequential Data and Time-Series Processing, Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), Attention Mechanism, Transformers and Large Language Models (Overview), Applications of Deep Learning in NLP, Healthcare, Industry 4.0, and IoT.	[8]
Text Books [T1] Goodfellow, I., Bengio, Y., Courville, A., & Bengio, Y. (2016). Deep learning (1st ed.). Cambridge: MIT press. [T2] Aggarwal, C. C. (2023). Neural networks and deep learning (2nd ed.). Cham: Springer.	
Reference Books [R1] Chollet, F., & Chollet, F. (2021). Deep learning with Python (2nd ed.). simon and schuster. [R2] Zhang, A., Lipton, Z. C., Li, M., & Smola, A. J. (2023). Dive into deep learning (1st ed.). Cambridge University Press.	

Paper code: ARI373T								L	P	Credit	
Paper Title: IoT Enabled Smart Grid and Energy Systems								3	0	3	
Marking Scheme:											
Teachers Continuous Evaluation: As per university examination norms from time to time.											
End Term Theory Examination: As per university examination norms from time to time.											
INSTRUCTIONS TO PAPER SETTERS: Maximum Marks: As per University norms											
➤ There should be 9 questions in the end term examination question paper. ➤ Question No. 1 should be compulsory and cover the entire syllabus. This question should have objective or short answer type questions. ➤ Apart from Question No. 1, the rest of the paper shall consist of four units As per the syllabus. Every unit should have two questions. However, students may be asked to attempt only 1 question from each unit. ➤ The questions are to be framed keeping in view the learning outcomes of course/paper. The standard/ level of the questions to be asked should be at the level of the prescribed textbooks. ➤ The requirement of (scientific) calculators/ log-tables/ data-tables may be specified if required											
Course Outcomes [Bloom’s Knowledge Level (KL)]:											
CO1: Explain the fundamentals and key components of Smart Grids, including smart metering, IoT-enabled energy management, grid monitoring, and optimization.											
CO2: Explain microgrid architecture, operation, and control strategies. [K2]											
CO3: Analyze Smart Grid energy management systems, power quality issues, and monitoring techniques for reliable grid operation. [K4]											
CO4: Evaluate Smart Grid security techniques and energy analytics for secure and reliable grid operation. [K5]											
CO/PO	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11
CO1	3	3	3	3	3	1	-	-	1	-	3
CO2	3	3	3	3	3	1	-	-	1	-	3
CO3	3	3	3	3	3	1	-	-	1	1	3
CO4	3	3	3	3	3	1	-	-	1	1	3
Course Content											No of Lectures
Unit I Introduction to Smart Grid: Fundamentals of smart grids, Difference between conventional and smart grid; Intelligent appliances, Home energy management systems; Smart metering infrastructure; load leveling using IoT enable appliances; Grid monitoring and substation automation. Grid optimization (Volt/VAR Control); Fault isolation and service restoration.											[8]
Unit II Microgrids Operation and Control: Microgrid Architecture: Single facility microgrids; Multiple facility microgrids; Feeder microgrids; Substation microgrids; Operation and Control of microgrids; Control Strategies Under Grid-Connected Operation; Control Strategy for an Islanded Microgrid.											[8]

Unit III Smart Grid Energy Management: Home energy management system, Meter data management system; Outage management; Power Quality and EMC in Smart Grid; Power Quality issues of Grid connected Renewable Energy Sources; Power Quality Conditioners for Smart Grid; Web based Power Quality monitoring.	[8]
Unit IV Smart Grid Security and Energy analytics: Intelligent Electronic Devices (IED) & their application for monitoring & security; Identify vulnerability in IoT devices used in smart grid; False data injection attack in smart grids; smart grid encryption standards; load forecasting using machine learning, renewable energy prediction; Predictive maintenance of the transformer.	[8]
Text Books [T1] Ekanayake, J., Liyanage, K., Wu, J., Yokoyama, A., Jenkins, N., & Lopes, J. A. P. (2021). Smart grid: Fundamentals, design, technology, applications, communication and security (2nd ed., Indian adaptation). Wiley. [T2] Ouahada, K., & Longe, O. M. (Eds.). (2020). Smart energy management for smart grids (1st ed.). MDPI.	
Reference Books [R1] Abulanwar, S., et al. (2026). Hybrid AC/DC microgrids: Optimization, energy management, and protection (1st ed.). Springer. [R2] Poonia, R. C., et al. (2026). Smart cyber-physical systems: Innovations and real-world implications (1st ed.). CRC Press/Taylor & Francis.	

Paper code: ARI375T								L	P	Credit	
Paper Title: Mobile Application Development for IoT								3	0	3	
Marking Scheme: Teachers Continuous Evaluation: As per university examination norms from time to time. End Term Theory Examination: As per university examination norms from time to time.											
INSTRUCTIONS TO PAPER SETTERS: Maximum Marks: As per University norms											
➤ There should be 9 questions in the end term examination question paper. ➤ Question No. 1 should be compulsory and cover the entire syllabus. This question should have objective or short answer type questions. ➤ Apart from Question No. 1, the rest of the paper shall consist of four units As per the syllabus. Every unit should have two questions. However, students may be asked to attempt only 1 question from each unit. ➤ The questions are to be framed keeping in view the learning outcomes of course/paper. The standard/ level of the questions to be asked should be at the level of the prescribed textbooks. ➤ The requirement of (scientific) calculators/ log-tables/ data-tables may be specified if required.											
Course Outcomes [Bloom’s Knowledge Level (KL)]:											
CO1: Understand the fundamentals of mobile application development and mobile platforms. [K1, K2]											
CO2: Develop native mobile applications using Android and iOS frameworks. [K2, K3]											
CO3: Develop cross-platform mobile applications using Flutter and React Native. [K2, K3, K4]											
CO4: Evaluate mobile applications for testing, deployment, security, and performance. [K3, K4, K5]											
CO/PO	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11
CO1	3	2	1	-	1	-	-	-	-	-	2
CO2	3	3	3	2	3	-	-	1	2	-	2
CO3	3	2	3	3	3	-	-	2	-	-	3
CO4	2	3	2	3	2	1	-	1	2	-	2
Course Content											No of Lectures
Unit-I Introduction to Mobile Application Development Introduction to mobile computing and mobile application development, Mobile device architecture (ARM and RISC-based processors), mobile operating systems, process isolation and application sandboxing, memory management, battery and power management, resource-constrained computing environments, Mobile Application type: native, hybrid and cross-platform applications. Mobile UI/UX principles, development environments, mobile security, privacy, performance considerations.											[8]
Unit-II Native Mobile Application Development Android architecture, Android application components: activity, intent, service, user interface											[8]

Approved by BoS of USAR : 16/6/2026

Approved by AC sub-committee : 20/07/2026

Applicable for Batch admitted in 2025 and Lateral Entry of AS 2026-27 admitted in Batch 2025

development, data storage, notification, location service. Android Studio environment, Java and Kotlin programming for Android applications. iOS architecture, application lifecycle, view controllers, user interface development, data persistence, notification, device service, Xcode environment, Objective-C and Swift programming for iOS applications. Development of basic native mobile applications.	
Unit-III Cross-Platform Mobile Application Development Flutter and Dart fundamentals, widgets, layouts, navigation, state management, local storage, API integration and Firebase services. React Native architecture, JavaScript/TypeScript fundamentals, components, styling, navigation, hooks, state management, local storage, API integration and device integration. Development of cross-platform mobile applications using Flutter and React Native.	[8]
Unit-IV Mobile Backend, Testing And Deployment RESTful APIs, cloud integration, authentication, notifications, location-based services, sensors and Firebase integration. Mobile application testing, debugging, performance optimization, mobile security, application permissions, secure data storage, mobile malware, rooting and jailbreaking concepts, version control, application packaging, signing, versioning, CI/CD concepts, deployment and publishing on Google Play Store and Apple App Store, privacy policies, end-user license agreements (EULA), app store compliance requirements, maintenance and emerging trends in mobile application development.	[8]
Text Books [T1] Griffiths, D., & Griffiths, D. (2024). Head first Android development (3rd ed.). O'Reilly Media. [T2] Ray, W. (2024). iOS apprentice: Beginning iOS development with Swift (9th ed.). Kodeco. [T3] Osmani, A. (2023). Learning Flutter: Building multiplatform apps with Dart, Flutter, and Firebase (2nd ed.). O'Reilly Media. [T4] Dabit, N. (2019). React Native in action (1st ed.). Manning Publications.	
Reference Books [R1] Phadke, A., & Kasam, M. (2022). Mobile application development (1st ed.). TechKnowledge Publications. [R2] Phillips, B., Stewart, C., Hardy, B., & Marsicano, K. (2022). Android programming: The Big Nerd Ranch guide (5th ed.). Big Nerd Ranch. [R3] Nahavandipoor, P. (2024). iOS 18 programming fundamentals with Swift: Swift, SwiftUI, and UIKit (1st ed.). O'Reilly Media. [R4] Fling, B. (2009). Mobile design and development: Practical concepts and techniques for creating mobile sites and web apps (1st ed.). O'Reilly Media.	

Paper code: ARI377T								L	P	Credit	
Paper Title: Electrical Vehicles and Drives								3	0	3	
Marking Scheme: Teachers Continuous Evaluation: As per university examination norms from time to time. End Term Theory Examination: As per university examination norms from time to time.											
INSTRUCTIONS TO PAPER SETTERS: Maximum Marks: As per University norms											
➤ There should be 9 questions in the end term examination question paper. ➤ Question No. 1 should be compulsory and cover the entire syllabus. This question should have objective or short answer type questions. ➤ Apart from Question No. 1, the rest of the paper shall consist of four units As per the syllabus. Every unit should have two questions. However, students may be asked to attempt only 1 question from each unit. ➤ The questions are to be framed keeping in view the learning outcomes of course/paper. The standard/ level of the questions to be asked should be at the level of the prescribed textbooks. ➤ The requirement of (scientific) calculators/ log-tables/ data-tables may be specified if required.											
Course Outcomes [Bloom’s Knowledge Level (KL)]: CO1: Understand the concept of electric vehicles (EVs) including motor selection and storage units. [K1, K2] CO2: Understand basics of power flow and their control analysis in Evs. [K1, K2, K3] CO3: Understand basic concepts, analyze and apply DC motor and induction motor drives in Evs. [K2, K3, K4] CO4: Understand the basic concepts of charging and energy management in Evs. [K1, K2].											
CO/PO	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11
CO1	3	2	1	-	-	2	-	-	-	-	2
CO2	3	3	2	1	2	-	-	-	-	-	2
CO3	3	3	3	1	3		-	-	-	-	1
CO4	3	2	2	1	-	3	1	1	-	2	3
Course Content											No of Lectures
Unit I Introduction to Electrical Vehicles (EV): EV development: past, present and future, comparison with IC engine driven vehicles; Power converters in EV; Different types of motors used in EV; Torque-speed characteristics of motors used in EV; motor control techniques; Storage units: Batteries, fuel cells, ultra-capacitors.											[8]
Unit II Power Flow and Control in EV: EV energy fundamentals; Powertrain architecture and voltage control; Pure EV power control – Mode 1 (Normal & Acceleration); Mode 2 (Light Load & Eco Modes); Mode 3 (Regenerative Braking) and Mode 4 (Vehicle Standstill); Hybrid advanced power control –Dual motor splitting; Multi-motor efficiency; Grid-to-Vehicle (G2V) and Vehicle-to-Grid (V2G) Flow.											[8]

Unit III Electric Vehicle Drives: DC Motor Drive and application Combined armature voltage and field control; Chopper control of DC motors; Multi quadrant control of chopper-fed DC motor drives; Class-c two quadrant chopper; four-quadrant operation; Application of DC motor drive. Electric Vehicle Drives: Induction Motor Drives and application Constant Volt/Hertz control; Field orientation control; Direct rotor flux orientation scheme; Indirect rotor flux orientation scheme; Voltage control in voltage source inverter; Current control in voltage source inverter; Application of Induction motor drive.	[8]
Unit IV Electric Vehicles Charging Station: Types of charging station; Selection and sizing of charging station; Components of charging station; Single line diagram of charging station; Charging station placement for electric vehicles; Different energy management strategies in electric vehicles, Battery management and safety limits.	[8]
Textbooks: [T1] Ehsani, M., Gao, Y., Gay, S. E., & Emadi, A. (2005). Modern electric, hybrid electric, and fuel cell vehicles: Fundamentals, theory, and design (1st ed.). CRC Press. [T2] Enge, P., Enge, N., & Zoepf, S. (2021). Electric vehicle engineering (1st ed.). McGraw Hill.	
Reference Books: [R1] Larminie, J., & Lowry, J. (2003). Electric vehicle technology explained (1st ed.). John Wiley & Sons. [R2] Garcia-Valle, R., & Peças Lopes, J. A. (2013). Electric vehicle integration into modern power networks (1st ed.). Springer.	

DETAILED SYLLABUS FOR 6th SEMESTER

Paper code: ARI302								L	P	Credit	
Paper Title: Embedded Systems								3	0	3	
Marking Scheme: Teachers Continuous Evaluation: As per university examination norms from time to time. End Term Theory Examination: As per university examination norms from time to time.											
INSTRUCTIONS TO PAPER SETTERS: Maximum Marks: As per University norms											
➤ There should be 9 questions in the end term examination question paper. ➤ Question No. 1 should be compulsory and cover the entire syllabus. This question should have objective or short answer type questions. ➤ Apart from Question No. 1, the rest of the paper shall consist of four units As per the syllabus. Every unit should have two questions. However, students may be asked to attempt only 1 question from each unit. ➤ The questions are to be framed keeping in view the learning outcomes of course/paper. The standard/level of the questions to be asked should be at the level of the prescribed textbooks. ➤ The requirement of (scientific) calculators/ log-tables/ data-tables may be specified if required.											
Course Outcomes [Bloom’s Knowledge Level (KL)]: CO1: Students will Understand the importance, use and architecture of Microcontrollers. [K1, K2, K3] CO2: Ability of students to understand the AVR Microcontroller and apply the knowledge for its detailed configuration. [K1, K2, K3] CO3: Analyze the various modes and timers of AVR microcontroller along and evaluate output with the C program of AVR. [K4, K5] CO4: Understand and Evalaute the interfacing of AVR microcontroller and apply it to various electronic and electrical devices. [K2, K3, K4]											
CO/PO	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11
CO1	3	3	3	2	2	-	-	-	1	1	2
CO2	3	3	2	3	3	-	-	1	1	2	2
CO3	3	3	2	3	3	-	-	1	1	2	3
CO4	3	3	2	3	3	-	-	1	1	2	3
Course Content											No of Lectures
Unit-I Introduction of Embedded System: Overview of Embedded Systems, Features, Requirements and Applications, Recent Trends in the Embedded System Design, Common architectures for the Embedded System Design, Embedded Software design issues. Introduction to microcontrollers, Overview of Harvard architecture and Von Neumann architecture, RISC and CISC microcontrollers.											[7]

Unit-II Architecture & Assembly Instruction: Introduction to AVR Microcontroller, Architecture and Pin Configuration, Register and memory mapping, Status Register, Instruction set, Data Transfer Instructions, Arithmetic and Logic Instructions, Branch Instructions, Bit and Bit-test Instructions, MCU Control Instructions, Delay time loop.	[8]
Unit-III Interrupts and Timer of AVR: Introduction to System Clock, Reset sources, Introduction to interrupts, External interrupts, UART- Basic Operation, I/O Register configuring, I/O Ports, 8-bit and 16-bit Timer block diagram, Modes- Output Compare Mode, Fast PWM Mode, CTC Mode, Simple programs in C Language, AVR I/O Port Programming.	[8]
Unit IV Peripherals Interfacing with AVR: Analog Comparator, ADC, DAC and sensor interfacing, Serial Peripheral Interface (SPI), The Universal Synchronous and Asynchronous serial Receiver and Transmitter (USART), Two Wire Interface (TWI), I2C Protocol and RTC interfacing, 7-Segment LED Display, LCD and Keyboard Interfacing, Opto-isolator and Stepper Motor Interfacing, Relay.	[9]
Text Books [T1] Muhammad Ali Mazidi, Shujen Chen, Sepehr Naimi, Sarmad Naimi (2017), The AVR Microcontroller and Embedded Systems: Using Assembly and C(1st ed.). Freescale ARM Cortex-M. [T2] Dhananjay Gadre, (2017) Programming and Customizing the AVR Microcontroller (1st ed.). McGraw Hill, Education.	
Reference Books [R1] Noergaard, T. (2012). Embedded systems architecture: a comprehensive guide for engineers and programmers (2nd ed.). Newnes.	

Paper code: ARI304							L	P	Credit		
Paper Title: Data Analytics for IoT							3	0	3		
Marking Scheme: Teachers Continuous Evaluation: As per university examination norms from time to time. End Term Theory Examination: As per university examination norms from time to time.											
INSTRUCTIONS TO PAPER SETTERS: Maximum Marks: As per University norms											
➤ There should be 9 questions in the end term examination question paper. ➤ Question No. 1 should be compulsory and cover the entire syllabus. This question should have objective or short answer type questions. ➤ Apart from Question No. 1, the rest of the paper shall consist of four units As per the syllabus. Every unit should have two questions. However, students may be asked to attempt only 1 question from each unit. ➤ The questions are to be framed keeping in view the learning outcomes of course/paper. The standard/ level of the questions to be asked should be at the level of the prescribed textbooks. ➤ The requirement of (scientific) calculators/ log-tables/ data-tables may be specified if required											
Course Outcomes [Bloom’s Knowledge Level (KL)]: CO1: Understand IoT data, preprocessing, and visualization techniques. [K1, K2] CO2: Evaluate predictive analytics models using classification, ensemble learning, dimensionality reduction, and model optimization techniques for IoT applications. [K5] CO3: Apply regression, clustering, and model optimization techniques. [K2, K3, K4] CO4: Analyze big data and IoT analytics applications. [K4, K5]											
CO/PO	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11
CO1	3	2	-	-	1	-	-	-	-	-	2
CO2	3	3	2	2	3	-	-	1	-	1	2
CO3	3	2	3	2	3	-	-	1	-	1	2
CO4	2	3	2	3	2	-	-	1	-	2	3
Course Content											No of Lectures
Unit-I Data Analytics Fundamentals: Introduction to Data Analytics, Nature of data, Characteristics of IoT Data, Data Acquisition and Preparation, Data Preprocessing, Feature selection and Extraction, Feature Engineering, Exploratory Data Analysis, Descriptive Statistics, Time-Series Analysis, Data Visualization and Visual Analytics.											[7]
Unit-II Predictive Analytics for IoT: Classification problems, Artificial Neural Networks, Support Vector Machines, K-Nearest Neighbor, Naïve Bayes, Decision Tree classifiers, Ensemble Modeling Techniques (Voting, Stacking, Bagging (Random Forest), Boosting (AdaBoost, XGBoost)), Dimensionality Reduction (Principal Component Analysis (PCA), Linear Discriminant Analysis (LDA)), Hyperparameter Tuning, Cross Validation.											[8]

Approved by BoS of USAR : 16/6/2026

Approved by AC sub-committee : 20/07/2026

Applicable for Batch admitted in 2025 and Lateral Entry of AS 2026-27 admitted in Batch 2025

Unit-III Regression and Clustering: Linear regression, Lasso, Ridge, ElasticNet regression, Non-Linear regression, Cluster Analysis (K-means clustering, K-Means ++, Hierarchical clustering, DBSCAN, other clustering techniques.	[8]
Unit IV Big Data and Analytics: Big Data Fundamentals and Characteristics, The “V”s That Define Big Data, Hadoop Ecosystem and MapReduce, NoSQL Databases (Key-Value, Document and Column-Family Stores), Apache Spark and In-Memory Analytics, Stream Analytics and Data Stream Mining, Cloud-Based Analytics. Advanced IoT Analytics Applications: Essentials of IoT, IoT Ecosystem and Architecture, IoT and Decision Support, IoT Applications, Smart Cities and Smart Factories, Analytics and IoT Integration, Implementation Issues in IoT, Security and Privacy Issues, Future of IoT.	[9]
Text Books [T1] Sharda, R., Delen, D., & Turban, E. (2021). Analytics, data science, & artificial intelligence: Systems for decision support (11th Global ed.). Pearson Education Limited. [T2] Acharya, S., & Chellappan, S. (2019). Big data and analytics (2nd ed.). Wiley India Pvt. Ltd.	
Reference Books [R1] Han, J., Kamber, M., & Pei, J. (2012). Data mining: Concepts and techniques (3rd ed.). Morgan Kaufmann. [R2] McKinney, W. (2018). Python for data analysis: Data wrangling with pandas, NumPy, and Jupyter (2nd ed.). O'Reilly Media. [R3] Rajaraman, A., & Ullman, J. D. (2011). Mining of massive datasets (1st ed.). Cambridge University Press.	

Paper code: ARI306							L	P	Credit		
Paper Title: Cyber Security and Cryptography for Industrial Systems							3	0	3		
Marking Scheme: Teachers Continuous Evaluation: As per university examination norms from time to time. End Term Theory Examination: As per university examination norms from time to time.											
INSTRUCTIONS TO PAPER SETTERS: Maximum Marks: As per University norms											
➤ There should be 9 questions in the end term examination question paper. ➤ Question No. 1 should be compulsory and cover the entire syllabus. This question should have objective or short answer type questions. ➤ Apart from Question No. 1, the rest of the paper shall consist of four units As per the syllabus. Every unit should have two questions. However, students may be asked to attempt only 1 question from each unit. ➤ The questions are to be framed keeping in view the learning outcomes of course/paper. The standard/level of the questions to be asked should be at the level of the prescribed textbooks. ➤ The requirement of (scientific) calculators/ log-tables/ data-tables may be specified if required.											
Course Outcomes [Bloom’s Knowledge Level (KL)]:											
CO1: Understand cyber security fundamentals, risk, threats, vulnerabilities, and attack techniques in industrial systems. [K1, K2]											
CO2: Analyze industrial networks, protocols, and security mechanisms. [K2, K3]											
CO3: Apply risk management, incident response, and digital forensic techniques. [K3, K4]											
CO4: Analyze cryptographic and lightweight security techniques for industrial systems. [K3, K4, K5]											
CO/PO	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11
CO1	3	2	-	-	-	-	-	-	-	-	2
CO2	3	3	2	2	2	-	-	1	-	-	2
CO3	2	3	2	2	3	-	-	1	1	-	2
CO4	3	2	2	2	3	2	-	1	1	-	3
Course Content											No of Lectures
Unit I: Cyber Security Fundamentals Cyber Security Objectives, Confidentiality, Integrity and Availability (CIA) Triad, Assets and Attack Surface, Authentication and Authorization, Risk, Threat and Vulnerability Concepts. Common Known Attack Types: Malware, Ransomware, Phishing, Social Engineering, Denial-of-Service (DoS), Distributed Denial-of-Service (DDoS), Man-in-the-Middle (MitM) Attacks. Common Vulnerabilities: Open Worldwide Application Security Project (OWASP) Top 10.											[7]
Unit-II Industrial Network and System Security Protocol Vulnerabilities, Network Segmentation and Zoning, Security Zones and Conduits, Secure Remote Access, Virtual Private Networks (VPNs), Industrial Firewalls, Intrusion Detection and Prevention Systems (IDS/IPS), Access Control Mechanisms, Network Monitoring and Logging.											[7]

Approved by BoS of USAR : 16/6/2026

Approved by AC sub-committee : 20/07/2026

Applicable for Batch admitted in 2025 and Lateral Entry of AS 2026-27 admitted in Batch 2025

Unit-III Risk Management and Digital Forensics: Risk Assessment, Threat Modeling (STRIDE), Risk Mitigation Strategies, Security Monitoring, Anomaly Detection, Incident Detection and Response, Computer Security Incident Response Team (CSIRT), ISO/IEC 27001 Overview, Compliance and Security Best Practices; Digital Forensics Fundamentals, Evidence Collection and Preservation, Chain of Custody, Cyber Forensic Investigation Process, Data Protection and Privacy. Case Study: Reverse Engineering and Analysis of a Cyber Attack on Critical Infrastructure.	[9]
Unit-IV Cryptography and Lightweight Security Cryptography Fundamentals, Advanced Encryption Standard (AES), Rivest-Shamir-Adleman (RSA), Cryptographic Hash Functions (SHA-256); Lightweight Cryptography Fundamentals, Resource-Constrained Devices, Lightweight Security Requirements, Lightweight Hash Functions (PHOTON), Lightweight MACs (SipHash), Impact of Quantum Computing on Classical Cryptographic Systems.	[9]
Text Books [T1] Stallings, W., & Brown, L. (2024). Computer security: Principles and practice (5th ed.). Pearson. [T2] Ackerman, P. (2024). Industrial cybersecurity: Efficiently secure critical infrastructure systems (3rd ed.). Packt Publishing. [T3] Shostack, A. (2014). Threat modeling: Designing for security (1st ed.). Wiley. [T4] Nelson, B., Phillips, A., & Steuart, C. (2024). Guide to computer forensics and investigations (7th ed.). Cengage Learning.	
Reference Books [R1] NIST. (2023). Guide to operational technology (OT) security (NIST SP 800-82 Rev. 3). National Institute of Standards and Technology. [R2] Casey, E. (2024). Digital evidence and computer crime (4th ed.). Academic Press. [R3] ISO/IEC. (2022). ISO/IEC 27001:2022 Information security, cybersecurity and privacy protection: Information security management systems requirements (3rd ed.). International Organization for Standardization (ISO). [R4] Turan, M. S., McKay, K. A., Chang, D., Kang, J., & Kelsey, J. (2025). Ascon-based lightweight cryptography standards for constrained devices: Authenticated encryption, hash, and extendable output functions (NIST SP 800-232). National Institute of Standards and Technology.	

Paper code: ARI308								L	P	Credit	
Paper Title: Cloud, Dew, Edge and Fog Computing								3	0	3	
Marking Scheme: Teachers Continuous Evaluation: As per university examination norms from time to time. End Term Theory Examination: As per university examination norms from time to time.											
INSTRUCTIONS TO PAPER SETTERS: Maximum Marks: As per University norms											
➤ There should be 9 questions in the end term examination question paper. ➤ Question No. 1 should be compulsory and cover the entire syllabus. This question should have objective or short answer type questions. ➤ Apart from Question No. 1, the rest of the paper shall consist of four units As per the syllabus. Every unit should have two questions. However, students may be asked to attempt only 1 question from each unit. ➤ The questions are to be framed keeping in view the learning outcomes of course/paper. The standard/ level of the questions to be asked should be at the level of the prescribed textbooks. ➤ The requirement of (scientific) calculators/ log-tables/ data-tables may be specified if required.											
Course Outcomes [Bloom’s Knowledge Level (KL)]:											
CO1: Understand cloud computing architectures, services, and deployment models. [K1, K2]											
CO2: Apply virtualization and cloud service technologies. [K2, K3]											
CO3: Analyze cloud, fog, and edge computing paradigms for IoT. [K3, K4]											
CO4: Apply fog and edge computing techniques using analytics and simulation tools. [K3, K4, K5]											
CO/PO	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11
CO1	3	3	3	3	2	-	-	1	2	-	3
CO2	3	3	3	3	2	-	-	1	3	-	3
CO3	3	3	3	3	2	-	-	1	2	-	3
CO4	3	3	3	3	2	-	-	1	3	-	3
Course Content										No of Lectures	
Unit-I Cloud Computing Fundamentals Introduction to Cloud Computing, Definition, Characteristics, Components, Cloud Service Providers, Service Models (Software as a Service (SaaS), Platform as a Service (PaaS), Infrastructure as a Service (IaaS)), Deployment Models (Public, Private, Hybrid and Community Cloud), Resource Provisioning, Load Balancing, Resource Optimization, Cloud Architecture, Cloud Platforms and Comparative Study of Amazon EC2, Microsoft Azure, Google Cloud Platform and MeghRaj										[8]	
Unit-II Cloud Enabling Technologies and Virtualization Cloud Enabling Technologies, Hypervisors and Virtualization, Virtual Machine Management, Containerization, Web Services, Simple Object Access Protocol (SOAP), Representational State Transfer (REST), SOAP vs REST, Mashups and Web APIs, Multi-Tenant Architectures, Multi-Tenancy using Cloud Data Stores, Cloud Storage, Sensor-Cloud Integration, Workflow Management and Workflow Execution in Cloud Environments.										[8]	

Approved by BoS of USAR : 16/6/2026

Approved by AC sub-committee : 20/07/2026

Applicable for Batch admitted in 2025 and Lateral Entry of AS 2026-27 admitted in Batch 2025

Unit-III Security Issues in IoT Cloud Security Fundamentals, Cloud Security Architecture, Security Threats and Attacks in Cloud Computing, Privacy and Data Protection, Vulnerability Assessment in Cloud Environments, Quality of Service (QoS), Service Level Agreement (SLA), Inter-Cloud and Federated Cloud Issues.	[8]
Unit-IV Fog and Edge Computing & Emerging Trends Fog Computing, Edge Computing, IoT-Fog-Cloud Integration, Resource Management and Optimization in Fog and Edge Environments. Mist Computing, Dew Computing, Fog Middleware, Edge Architectures, Data Management and Analytics, Edge Analytics, Fog Orchestration, Generative Artificial Intelligence (Generative AI) Services and Agentic Artificial Intelligence (Agentic AI) in Cloud Environments. Case Study: Design and Development of MiCEF (Mist-IoT-Cloud-Edge-Fog) Computing Applications using CloudSim and iFogSim.	[8]
Text Books [T1] Buyya, R., Broberg, J., & Goscinski, A. (Eds.). (2024). Cloud computing: Principles and paradigms. John Wiley & Sons (2nd ed.). [T2] Erl, T., Mahmood, Z., & Puttini, R. (2023). Cloud computing: Concepts, technology & architecture (2nd ed.). Pearson.	
Reference Books [R1] Sosinsky, B. (2011). Cloud computing bible (1st ed.). Wiley. [R2] Buyya, R., & Srirama, S. N. (Eds.). (2019). Fog and edge computing: Principles and paradigms (1st ed.). John Wiley & Sons. [R3] Varghese, B., & Buyya, R. (2024). Next generation cloud computing: New trends and research directions (1st ed.). CRC Press. [R4] Alva, L. R., & Pandey, B. (2026). Agentic AI systems in the age of generative models: Architectures, cloud scalability, and real-world applications. Artificial Intelligence Review.	

Paper Code: ARI372T								L	P	Credit	
Paper Title: Fundamentals of Robotics								3	0	3	
Marking Scheme: Teachers Continuous Evaluation: As per university examination norms from time to time. End Term Theory Examination: As per university examination norms from time to time.											
INSTRUCTIONS TO PAPER SETTERS: Maximum Marks : As per University norms											
➤ There should be 9 questions in the end term examination question paper. ➤ Question No. 1 should be compulsory and cover the entire syllabus. This question should have objective or short answer type questions. ➤ Apart from Question No. 1, the rest of the paper shall consist of four units As per the syllabus. Every unit should have two questions. However, students may be asked to attempt only 1 question from each unit. ➤ The questions are to be framed keeping in view the learning outcomes of course/paper. The standard/level of the questions to be asked should be at the level of the prescribed textbooks. ➤ The requirement of (scientific) calculators/ log-tables/ data-tables may be specified if required.											
Course Outcomes [Bloom’s Knowledge Level (KL)]: CO1: Explain the fundamental concepts of robotics, robot components, configurations, degrees of freedom, workspace, and applications across various industrial sectors [K2, K3]. CO2: Analyze robot kinematics, coordinate transformations, dynamics, trajectory planning, and control techniques for robotic manipulators [K4]. CO3: Classify different types of robots and describe robot programming concepts, programming languages, drive systems, sensors, actuators, and industrial robotic architectures [K2, K3]. CO4: Apply the principles of intelligent robotics, artificial intelligence, machine learning, computer vision, and human–robot collaboration to solve real-world robotic problems [K3, K4].											
CO/PO	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11
CO1	2	2	2	2	2	2	-	1	-	1	3
CO2	2	3	3	3	3	2	-	1	-	1	3
CO3	3	3	3	3	3	2	-	1	-	-	3
CO4	3	2	2	2	3	2	-	1	-	-	3
Course Content											No of Lectures
Unit-I Introduction: Fundamentals of Robotics, Robot Kinematics: Position Analysis, Dynamic Analysis and Forces. Components of Robotic Systems such as Sensors, Actuators, Controllers, End-Effectors, and Power Systems; Degrees of Freedom (DOF), Robot Configurations, Workspace, and Performance Characteristics; Robot Programming Concepts; Applications of Robotics in Manufacturing, Healthcare, Agriculture, Logistics, Defense.											[9]

Unit-II Industrial Robots: Robot Classification, including Industrial, Service, Mobile, Humanoid, and Collaborative Robots; medical robots, Robot Specification, kinematic representations and transformations, dynamics techniques; trajectory planning and control. Robot Programming languages & systems: Introduction, the three levels of robot programming, requirements of a robot programming language, problems peculiar to robot programming languages.	[8]
Unit-III Robotics and Its applications: DDD concept, Intelligent robots, Robot anatomy-Definition, law of robotics, History and Terminology of Robotics-Accuracy and repeatability of Robotics-Simple problems-Specifications of Robot-Speed of Robot, Robot joints and links-Robot classifications Architecture of robotic systems-Robot Drive systems-Hydraulic, Pneumatic and Electric system.	[8]
Unit-IV Introduction AI&ML in Robotics: Supervised, Unsupervised, and Reinforcement Learning Techniques; Deep Learning for Robotic Perception and Decision-Making; Computer Vision and Natural Language Processing Applications in Robotics; Human–Robot Interaction and Collaborative Robotics; Cloud Robotics, Edge AI, and Digital Twin Technologies; Swarm Robotics and Multi-Agent Systems; Generative AI and Foundation Models for Robotics; Ethical, Safety, Security, and Future Trends in Intelligent Robotic Systems.	[7]
Text Books [T1] Niku, S. B. (2020). Introduction to Robotics: Analysis, Control, and Applications (3rd ed.). Wiley. [T2] Groover, M. P. (2020). Automation, Production Systems, and Computer-Integrated Manufacturing (5th ed.). Pearson. [T3] Kamal, R. (2024). Internet of Things: Architecture and Design Principles (3rd ed.). McGraw Hill Education. [T4] Gilchrist, A. (2025). Industry 4.0: The Industrial Internet of Things (2nd ed.). Apress.	
Reference Books [R1] Craig, J. J. (2018). Introduction to Robotics: Mechanics and Control (4th ed.). Pearson. [R2] Siciliano, B., Sciavicco, L., Villani, L., & Oriolo, G. (2022). Robotics: Modelling, Planning and Control. Springer. (Reprint edition). [R3] Lamb, F. (2021). Industrial Automation: Hands-On (2nd ed.). McGraw Hill Education. [R4] Uckelmann, D., Harrison, M., & Michahelles, F. (Eds.). (2021). Architecting the Internet of Things. Springer. (Reprint edition).	

Paper Code: ARI374T								L	P	Credit	
Paper Title: Introduction to Unmanned Systems								3	-	3	
Marking Scheme: Teachers Continuous Evaluation: As per university examination norms from time to time. End Term Theory Examination: As per university examination norms from time to time.											
INSTRUCTIONS TO PAPER SETTERS: Maximum Marks: As per University norms											
➤ There should be 9 questions in the end term examination question paper. ➤ Question No. 1 should be compulsory and cover the entire syllabus. This question should have objective or short answer type questions. ➤ Apart from Question No. 1, the rest of the paper shall consist of four units As per the syllabus. Every unit should have two questions. However, students may be asked to attempt only 1 question from each unit. ➤ The questions are to be framed keeping in view the learning outcomes of course/paper. The standard/ level of the questions to be asked should be at the level of the prescribed textbooks. ➤ The requirement of (scientific) calculators/ log-tables/ data-tables may be specified if required.											
Course Outcomes [Bloom’s Knowledge Level (KL)]: CO1: Explain the classification, architecture, and applications of Unmanned Aerial Systems. [K2] CO2: Analyze aerodynamics, flight mechanics, and hardware architecture of UAVs. [K4] CO3: Describe navigation, guidance, control, and communication systems used in UAS. [K3] CO4: Apply sensor integration, payload design, and regulatory knowledge to real-world UAS problems. [K3]											
CO/PO	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11
CO1	3	2	1	1	2	1	1	1	-	1	1
CO2	3	3	2	2	2	1	1	1	-	1	1
CO3	3	3	2	2	3	1	1	1	-	1	2
CO4	2	2	3	3	3	2	2	1	-	1	2
Course Content											No of Lectures
Unit 1 Introduction to Unmanned Aerial Systems (UAS): Evolution and historical development of Unmanned Aerial Vehicles (UAVs) / Drones, Definition, elements, and architecture of Unmanned Aerial Systems (UAS), Classification of Unmanned Aerial Vehicles (UAVs) / Drones based on size, range, endurance, and application, Types of Unmanned Aerial Vehicle (UAV) platforms: Multirotor, Fixed-wing, Vertical Takeoff and Landing (VTOL), Hybrid systems, Ground Control Stations and mission planning software Overview of civilian, commercial, and defense applications of UAS, Challenges, limitations, and future trends in UAS technology.											[8]

Unit II Aerodynamics, Flight Mechanics, and Hardware Architecture (Limited to Multi-Rotors): Basics of aerodynamics: Lift, drag, thrust, and weight, Thrust generation and control in multirotor UAVs, UAV structural design and materials. Propulsion systems: Motors, ESCs, propellers, Power systems: Batteries, power distribution, and safety considerations.	[8]
Unit III Navigation, Guidance, Control, and Communication Systems: Coordinate systems and reference frames, Sensor fusion and state estimation (Extended Kalman Filter (EKF) overview), Attitude, altitude, and position control loops (PID Controller), Waypoint navigation and autonomous flight modes, Radio Control (RC) communication systems and telemetry links, Communication protocols: MAVLink, Universal Asynchronous Receiver/Transmitter (UART), Inter-Integrated Circuit (I2C), Serial Peripheral Interface (SPI), Controller Area Network (CAN) (DroneCAN).	[9]
Unit-IV Sensors, Payloads, Regulations, and Emerging Trends: Imaging sensors: Red Green Blue (RGB), Electro-Optical (EO), Infrared (IR), Thermal, Range sensors (Time of Flight (TOF): LiDAR, ultrasonic, Radio Detection And Ranging (RADAR). Safety Regulations (UAS regulations): Directorate General of Civil Aviation (DGCA), overview: Airspace classification, permissions, and safety procedures, Ethical, privacy, and security consideration.	[7]
Text Books [T1] Austin, R. (2010). Unmanned aircraft systems: UAVs design, development and deployment (1st ed.). John Wiley & Sons. [T2] Beard, R. W., & McLain, T. W. (2012). Small unmanned aircraft: Theory and practice (1st ed.). Princeton University Press. [T3] Valavanis, K. P. (Ed.). (2018). Handbook of unmanned aerial vehicles (1st ed.). Springer.	
Reference Books [R1] Etkin, B., & Reid, L. D. (1996). Dynamics of flight: Stability and control (3rd ed.). John Wiley & Sons. [R2] Siegwart, R., Nourbakhsh, I. R., & Scaramuzza, D. (2011). Introduction to autonomous mobile robots (2nd ed.). MIT Press.	

Paper code: ARI376								L	P	Credit	
Paper Title: Metaverse and Blockchain								3	0	3	
Marking Scheme: Teachers Continuous Evaluation: As per university examination norms from time to time. End Term Theory Examination: As per university examination norms from time to time.											
INSTRUCTIONS TO PAPER SETTERS: Maximum Marks : As per University norms											
➤ There should be 9 questions in the end term examination question paper. ➤ Question No. 1 should be compulsory and cover the entire syllabus. This question should have objective or short answer type questions. ➤ Apart from Question No. 1, the rest of the paper shall consist of four units As per the syllabus. Every unit should have two questions. However, students may be asked to attempt only 1 question from each unit. ➤ The questions are to be framed keeping in view the learning outcomes of course/paper. The standard/ level of the questions to be asked should be at the level of the prescribed textbooks. ➤ The requirement of (scientific) calculators/ log-tables/ data-tables may be specified if required.											
Course Outcomes [Bloom’s Knowledge Level (KL)]: CO1: To understand the basic concepts of Blockchain Technology and Metaverse. [K2] CO2: Analyze Web 3.0 technologies, decentralized applications, smart contracts, and governance models, and evaluate their integration with the Metaverse. [K2, K3] CO3: To analyze the working of Smart Contracts. [K4] CO4: To apply Blockchain concepts to design smart contracts and develop blockchain solutions for securing cryptocurrency information. [K3, K4]											
CO \ PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	1	1	2	1	1	1	-	-	2
CO2	2	3	2	1	3	1	1	1	-	-	2
CO3	2	3	2	1	3	1	1	1	-	-	3
CO4	1	1	1	1	2	3	3	1	-	-	3
Course Content											No of Lectures
Unit-I Unit I: Foundations of Blockchain Technology and Metaverse: Introduction to Blockchain, Definition of Blockchain, What is Blockchain?, How is Blockchain used?, Origin of Blockchain Data Storage in Blockchain, Applications of Blockchain, Advantages and Disadvantages of Blockchain, Types of Blockchains: Public Blockchain, Private Blockchain, Fundamental Pillars of Blockchain Technology. Introduction to Metaverse: Definition of Metaverse, History of Metaverse, Applications of Metaverse, Advantages and Challenges of Metaverse, Key Technologies for Metaverse: Blockchain, Augmented Reality (AR), Virtual Reality (VR), Artificial Intelligence (AI), 3D Reconstruction, Internet of Things (IoT), Types of Metaverse, When will the Metaverse arrive?, Meta businesses,											[8]

Approved by BoS of USAR : 16/6/2026

Approved by AC sub-committee : 20/07/2026

Applicable for Batch admitted in 2025 and Lateral Entry of AS 2026-27 admitted in Batch 2025

Metaverse winners and losers, Metaverse existence, Cryptocurrencies and NFTs.	
Unit-II Building the Metaverse: Networking, Computing, Virtual World Engines, Interoperability, Hardware, Payment Rails. Web 3.0 and Metaverse: Evolution of the Web, Web 1.0 to Web 3.0 Transition, Characteristics and Architecture of Web 3.0, Technologies involved in Web 3.0 – Blockchain, Smart Contracts, Decentralized Storage, Peer-to-Peer Networks, Role of Artificial Intelligence in Web 3.0, Decentralized Applications (DApps), Decentralized Finance (DeFi), Smart Contracts and Decentralized Autonomous Organizations (DAOs), Web 3.0 and Metaverse Integration, Governance, Security and Privacy Issues in Web 3.0 and Metaverse.	[8]
Unit-III Smart Contract and Consensus Algorithm: Understanding Solidity Language, Creating Smart Contracts, Applications of Smart Contracts, BOSCA (Blockchain-Oriented Smart Contract Agreement), Blockchain Application Development using Remix IDE and Solidity.	[8]
Unit-IV BLAST: BlockChain Algorithm for Secure Transaction. Proof of Work (PoW), Proof of Stake (PoS), Non-Fungible Tokens (NFTs), Merkle Tree Formation, BLAST – Blockchain Algorithm for Secure Transactions. Design, develop, and deploy a Smart Contract using Remix IDE; Design a Blockchain-based solution to secure Credit Card/Debit Card information or Cryptocurrency information using an authentic cryptocurrency dataset.	[8]
Text Books and Web Resources [T1] Ball, M. (2022). The metaverse: and how it will revolutionize everything (1st ed.). Liveright Publishing. [T2] Bashir, I. (2020). Mastering blockchain (3rd ed.). Packt Publishing Ltd. [Web link] https://www.youtube.com/@RAHULJOHARI7/videos	
Reference Books [R1] Mahankali, S. (2019). Blockchain: The Untold Story: From birth of Internet to future of Blockchain (1st ed.). BPB Publications.	

Paper code: ARI378								L	P	Credits	
Subject: Micro-Electro-Mechanical-Systems (MEMS)								3	0	3	
Marking Scheme: Teachers Continuous Evaluation: As per university examination norms from time to time. End Term Theory Examination: As per university examination norms from time to time.											
INSTRUCTIONS TO PAPER SETTERS: Maximum Marks: AS per University norms											
➤ There should be 9 questions in the end term examination question paper. ➤ Question No. 1 should be compulsory and cover the entire syllabus. This question should have objective or short answer type questions. ➤ Apart from Question No. 1, the rest of the paper shall consist of four units as per the syllabus. Every unit should have two questions. However, students may be asked to attempt only 1 question from each unit. ➤ The questions are to be framed keeping in view the learning outcomes of course/paper. The standard/level of the questions to be asked should be at the level of the prescribed textbooks. ➤ The requirement of (scientific) calculators/ log-tables/ data-tables may be specified if required.											
Course Outcomes [Bloom’s Knowledge Level (KL)]:											
Course Outcomes:											
CO1: Understand the fundamentals of MEMS, microfabrication techniques, and micro-physics concepts used in microsystems. [K1, K2, K3]											
CO2: Analyze the operating principles, characteristics, and applications of MEMS sensors. [K1, K2]											
CO3: Evaluate the design, working mechanism, and performance of MEMS actuators and applications. [K2, K3]											
CO4: Understanding packaging concepts and interference in Microsystem Packaging. [K2, K3, K4]											
CO/PO	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11
CO1	3	2	2	3	-	-	-	-	-	3	2
CO2	3	2	3	3	-	-	-	-	-	2	2
CO3	3	3	3	3	-	-	-	-	-	2	3
CO4	3	3	3	2	-	-	-	-	-	3	2
Course Content											No of lectures
Unit I Introduction: Overview of MEMS & Microsystems: Miniaturization & Microsystems. Basic principles of scaling, surface-to-volume effects, and system integration. Materials used in MEMS: silicon, polymers, metals, ceramics, and thin films. MEMS Fabrication: Cleanroom processes and lithography. Bulk micromachining and surface micromachining. Etching techniques: wet etching, dry etching, anisotropic and isotropic etching. Thin film deposition, masking, bonding, and sacrificial layer processes. Packaging issues, reliability, and test methods for MEMS devices.											[8]
Unit II Micro Sensors: Position Sensors: Capacitive Sensors, Linear Variable Differential Transformer (LVDT), Interferometric Sensors, STM Tips based, etc. Force and Pressure Sensors: Strain Gauges, Deflection Based: AFM, etc., Visual Force Sensing: Bending Imaging, etc., Capacitive Force/Tactile Sensors. Accelerometers, Gyroscopes, Chemical Sensors, Flow Sensors, etc.											[8]

Unit III Micro actuators: Piezoelectric Actuators: Bending Type- Unimorph and Bimorphs, Stack Type- Piezotubes, Thin-Film Type: ZnO, etc. films, Surface Acoustic Waves, PZT actuators as also integrated sensors. Electrostatic, Thermal, Ultrasonic, Electro/Magnetostrictive, and Shape Memory Alloy Based Actuators. Polymer Actuators, Dielectric Elastomers, Carbon Nanotube (CNT) Actuators. Biomolecular Motors.	[8]
Unit IV Microsystem Packaging: Overview of Mechanical Packaging of Microelectronics systems, Microsystem Packaging: Die-level Packaging, Device-level Packaging, System -Level Packaging, Interference in Microsystem Packaging, Essential Packaging Technologies: Die Preparation, Surface Bonding, Wire Bonding, Sealing, Three-Dimensional Packaging, Selection of Packaging Materials.	[8]
Text Books: [T1] Liu, C. (2012). Foundations of MEMS (2nd ed.). Pearson. [T2] Hsu, T.-R. (2017). MEMS and Microsystems: Design and Manufacture (2nd ed.). McGraw Hill Education. [T3] Varadan, V. K., Vinoy, K. J., & Gopalakrishnan, S. (2006). Smart Material Systems and MEMS: Design and Development Methodologies (1st ed.). John Wiley & Sons.	
Reference Books: [R1] Madou, M. J. (2018). Fundamentals of Microfabrication and Nanotechnology (4th ed.). CRC Press. [R2] Senturia, S. D. (2001). Microsystem Design (1st ed.). Kluwer Academic Publishers. [R3] Lyshevski, S. E. (2002). MEMS and NEMS: Systems, Devices, and Structures (1st ed.). CRC Press.	

DETAILED SYLLABUS FOR 7th SEMESTER

Paper code: ARI401								L	P	Credit	
Paper Title: FPGA Based System Design								3	0	3	
Marking Scheme: Teachers Continuous Evaluation: As per university examination norms from time to time. End Term Theory Examination: As per university examination norms from time to time.											
INSTRUCTIONS TO PAPER SETTERS: Maximum Marks: As per University norms											
➤ There should be 9 questions in the end term examination question paper. ➤ Question No. 1 should be compulsory and cover the entire syllabus. This question should have objective or short answer type questions. ➤ Apart from Question No. 1, the rest of the paper shall consist of four units As per the syllabus. Every unit should have two questions. However, students may be asked to attempt only 1 question from each unit. ➤ The questions are to be framed keeping in view the learning outcomes of course/paper. The standard/ level of the questions to be asked should be at the level of the prescribed textbooks. ➤ The requirement of (scientific) calculators/ log-tables/ data-tables may be specified if required.											
Course Outcomes [Bloom’s Knowledge Level (KL)]: CO1: Apply Verilog HDL coding styles for modeling digital systems. [K1, K2] CO2: Understand FPGA architectures, technologies, and vendor-specific features. [K1, K2] CO3: Design and model combinational and sequential circuits using Verilog. [K3, K4] CO4: Implement synchronous sequential circuits and system-level designs on FPGAs. [K3, K4, K5]											
CO/PO	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11
CO1	3	2	2	1	2	2	-	-	-	-	2
CO2	3	2	2	2	2	2	-	-	-	-	2
CO3	3	3	3	2	2	2	-	-	-	-	2
CO4	3	3	3	3	2	3	-	-	-	-	3
Course Content											No of Lectures
Unit I Verilog HDL Fundamentals: Lexical conventions, ports, and modules, Operators and gate-level modelling, System tasks and compiler directives, Test benches for verification, Dataflow modeling and behavioral-level modelling, Tasks and functions in Verilog.											[8]
Unit II FPGA Architectures and Technologies: Overview of FPGA architectural options: coarse vs. fine-grained structures, Vendor-specific considerations (focus on Xilinx FPGAs), FPGA technologies: Antifuse, SRAM, and EPROM-based devices, FPGA logic cells, interconnection networks, and I/O pads.											[8]

Unit III Verilog Modeling of Circuits: Behavioral, dataflow, and structural realization of digital circuits, Carry Lookahead, Booth multiplier, Magnitude and Floating-Point comparators, Sequential circuits: flip-flops, shift registers, counters, Advanced components: synchronous/asynchronous FIFO, single-port RAM, Error detection with Cyclic Redundancy Check (CRC).	[8]
Unit IV Sequential Circuit Design and System Applications: State diagrams, state tables, and state assignments, Flip-flop choices and timing diagrams, One-hot encoding, Mealy and Moore state machines, Design examples: sequence detectors, System-level design using Xilinx FPGAs: traffic light controller, real-time clock, FPGA-based interfacing: VGA, keyboard, LCD, embedded processor hardware design.	[8]
Text Books [T1] Palnitkar, S. (2003). Verilog HDL: A Guide to Digital Design and Synthesis (2nd ed.). Prentice Hall. [T2] Ashenden, P. J. (2007). Digital Design Using VHDL (3rd ed.). Elsevier. [T3] Roy, S. (2024). Advanced Digital System Design: A Practical Guide to Verilog-Based FPGA and ASIC Implementation (1st ed.). Springer. [T4] Ashenden, P. J. (2007). Digital Design Using Verilog (1st ed.). Elsevier.	
Reference Books [R1] Wolf, W. (2004). FPGA-Based System Design (1st ed.). Pearson Education. [R1] Maxfield, C. (2004). The Design Warrior's Guide to FPGAs: Devices, Tools and Flows (1st ed.). Elsevier.	

Paper code: ARI403							L	P	Credit		
Paper Title: Industrial Photonics							3	0	3		
Marking Scheme:											
Teachers Continuous Evaluation: As per university examination norms from time to time.											
End Term Theory Examination: As per university examination norms from time to time.											
INSTRUCTIONS TO PAPER SETTERS: Maximum Marks: As per University norms											
➤ There should be 9 questions in the end term examination question paper. ➤ Question No. 1 should be compulsory and cover the entire syllabus. This question should have objective or short answer type questions. ➤ Apart from Question No. 1, the rest of the paper shall consist of four units As per the syllabus. Every unit should have two questions. However, students may be asked to attempt only 1 question from each unit. ➤ The questions are to be framed keeping in view the learning outcomes of course/paper. The standard/level of the questions to be asked should be at the level of the prescribed textbooks. ➤ The requirement of (scientific) calculators/ log-tables/ data-tables may be specified if required.											
Course Outcomes [Bloom’s Knowledge Level (KL)]:											
CO1: Understand the fundamental operating principles of industrial lasers, optical fibers, and optoelectronic components. [K1, K2, K3]											
CO2: Select and interface appropriate optical/fiber-optic sensors for real-time data acquisition in industrial IoT environments. [K1, K2, K3]											
CO3: Analyze the transmission, amplification for automated inspection and quality control. [K4, K5]											
CO4: Evaluate optical communication technologies (like Li-Fi and fiber networks) for robust, interference-free data transmission in smart factories. [K2, K3]											
CO/PO	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11
CO1	3	3	3	-	-	-	-	-	1	-	2
CO2	3	3	2	-	-	-	-	1	1	-	2
CO3	3	3	2	-	-	-	-	1	1	-	3
CO4	3	3	2	-	-	-	-	1	1	-	3
Course Content										No of Lectures	
Unit-I Fundamentals of Optoelectronics and Lasers Introduction to Photonics: Nature of light, polarization, interference, diffraction. total internal reflection, numerical aperture, step-index and graded-index fibers Laser Fundamentals: Spontaneous and stimulated emission, population inversion, optical resonators, and laser pumping. Types of Lasers in Industry: Semiconductor lasers, Nd:YAG, CO2, and Fiber lasers. Optoelectronic Devices: Principles and characteristics of LEDs, Photodiodes (PIN and Avalanche), Phototransistors, and Opto-isolators/Optocouplers.										[8]	

Unit-II Optical Sensing and IIoT Integration: Principles of Optical Sensors: Intensity-based, phase-based, and wavelength-based sensing. Fiber Optic Sensors: Fiber Bragg Grating (FBG) sensors for strain and temperature measurement. Advanced Industrial Sensors: LIDAR (Light Detection and Ranging) principles, Time-of-Flight (ToF) sensors, and laser distance meters for AGVs (Automated Guided Vehicles) and robotics. IoT Sensor Integration: Signal conditioning for optical sensors, interfacing photodetectors with Edge devices, and optical data transmission protocols.	[8]
Unit-III Optical amplification & Optical Inspection Optical Amplification: Fundamental receiver operation, Pre-amplifier types, Optical amplifiers, Semiconductor optical amplifiers, Erbium-doped fiber amplifiers, Amplifier noise, System applications. Industrial Applications: Automated Optical Inspection (AOI) for PCBs, barcode/QR code reading, surface defect detection, and dimensional measurement using optical metrology.	[8]
Unit IV Photonics in Industrial Communication and Manufacturing Optical Communication in IIoT application: Passive Optical Networks (PON) on the factory floor, plastic optical fibers (POF) for harsh environments. Wireless Optical Communication: Free Space Optics (FSO) and Li-Fi (Light Fidelity) applications, high-bandwidth communication in smart factories. Safety in Industrial Photonics: Laser safety classifications, eye protection, and international safety standards (IEC 60825) for industrial environments.	[8]
Text Books [T1] Kasap, S. O. (2024). Optoelectronics and Photonics: Principles and Practices (3rd ed.). Pearson. [T2] Ghatak, A., & Thyagarajan, K. (2022). Introduction to Fiber Optics (Updated Indian Edition). Cambridge University Press. [T3] Sharan, P., Roy, S. K., Patil, H. J., Chaudhary, A., & Kumar, D. (2026). Photonics in Industrial IoT: Transforming Manufacturing and Beyond: The Role of IoT Innovation in Smart Factories and Connected Environments (1st ed.). Apress. [T4] Keiser, G. (2020). Optical Fiber Communications (6th ed.). McGraw-Hill.	
Reference Books [R1] Ghatak, A., & Thyagarajan, K. (2023). Optical Electronics (1st ed.). Cambridge University Press. [R2] Singh, D. K. (2025). Free Space Optics: A Technology on Verge (1st ed.). LAP Lambert Academic Publishing. [R3] Udd, E., & Spillman Jr, W. B. (Eds.). (2011). Fiber Optic Sensors: An Introduction for Engineers and Scientists (2nd ed.). John Wiley & Sons.	

Paper code: ARI405								L	P	Credit	
Paper Title: Real Time Operating Systems								3	0	3	
Marking Scheme:											
Teachers Continuous Evaluation: As per university examination norms from time to time.											
End Term Theory Examination: As per university examination norms from time to time.											
INSTRUCTIONS TO PAPER SETTERS: Maximum Marks: As per University norms											
➤ There should be 9 questions in the end term examination question paper. ➤ Question No. 1 should be compulsory and cover the entire syllabus. This question should have objective or short answer type questions. ➤ Apart from Question No. 1, the rest of the paper shall consist of four units As per the syllabus. Every unit should have two questions. However, students may be asked to attempt only 1 question from each unit. ➤ The questions are to be framed keeping in view the learning outcomes of course/paper. The standard/ level of the questions to be asked should be at the level of the prescribed textbooks. ➤ The requirement of (scientific) calculators/ log-tables/ data-tables may be specified if required.											
Course Outcomes [Bloom’s Knowledge Level (KL)]:											
CO1: Explain the fundamentals, architecture, and characteristics of real-time operating systems. [K2]											
CO2: Apply scheduling and synchronization techniques in real-time systems. [K3]											
CO3: Analyze communication and memory management mechanisms in RTOS. [K4]											
CO4: Apply RTOS concepts for the design of embedded and real-time applications. [K3]											
CO/PO	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11
CO1	3	2	-	-	1	-	-	-	-	-	1
CO2	3	3	2	2	3	-	-	-	-	-	1
CO3	3	3	3	2	3	-	-	-	-	-	1
CO4	3	3	2	2	3	-	-	-	-	-	1
Course Content										No of Lectures	
Unit-I Fundamentals of Real-Time Systems: Introduction to Real-Time Systems, Characteristics and Applications of RTOS, Hard, Soft and Firm Real-Time Systems, Embedded Systems and RTOS, RTOS Architecture and Components, Tasks, Processes and Threads, Real-Time Kernel Concepts, Context Switching, Interrupt Handling and Interrupt Latency.										[8]	
Unit-II Task Scheduling and Resource Management: Real-Time Task Model, Task States and Task Control Blocks, Preemptive and Non-Preemptive Scheduling, Rate Monotonic Scheduling (RMS), Earliest Deadline First (EDF), Priority-Based Scheduling, Resource Sharing and Synchronization, Semaphores, Mutexes and Event Flags, Priority Inversion and Priority Inheritance.										[8]	

Unit-III: Interprocess Communication and Memory Management: Interprocess Communication (IPC), Message Queues, Mailboxes and Pipes, Shared Memory, Signals and Events, Memory Management in RTOS, Dynamic and Static Memory Allocation, Real-Time Memory Requirements, Deadlocks and Deadlock Handling.	[8]
Unit-IV RTOS Design and Applications: RTOS Performance Metrics, Reliability and Fault Tolerance, Real-Time System Design Considerations, Overview of Popular RTOS: FreeRTOS, VxWorks, QNX, RTOS for IoT and Industrial Automation, Case Studies and Applications in Embedded Systems.	[8]
Text Books [T1] Mall, R. (2009). Real-time systems: theory and practice (1st ed.). Pearson Education India. [T2] Liu, J. W. (2006). Real-time systems (1st ed.). Pearson Education India.	
Reference Books [R1] Li, Q., & Yao, C. (2003). Real-time concepts for embedded systems (1st ed.). CRC press.	

Paper code: ARI471								L	P	Credit	
Paper Title: Information Theory and Coding								3	0	3	
Marking Scheme: Teachers Continuous Evaluation: As per university examination norms from time to time. End Term Theory Examination: As per university examination norms from time to time.											
INSTRUCTIONS TO PAPER SETTERS: Maximum Marks: As per University norms											
➤ There should be 9 questions in the end term examination question paper. ➤ Question No. 1 should be compulsory and cover the entire syllabus. This question should have objective or short answer type questions. ➤ Apart from Question No. 1, the rest of the paper shall consist of four units As per the syllabus. Every unit should have two questions. However, students may be asked to attempt only 1 question from each unit. ➤ The questions are to be framed keeping in view the learning outcomes of course/paper. The standard/ level of the questions to be asked should be at the level of the prescribed textbooks. ➤ The requirement of (scientific) calculators/ log-tables/ data-tables may be specified if required.											
Course Outcomes [Bloom’s Knowledge Level (KL)]:											
CO1: Understand information theory and source coding principles. [K1, K2]											
CO2: Analyze channel coding concepts and communication limits. [K2, K3]											
CO3: Apply error control coding techniques for reliable communication. [K2, K3]											
CO4: Apply data compression techniques in communication systems. [K2, K3, K4]											
CO/PO	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11
CO1	3	2	-	-	-	-	-	-	-	-	2
CO2	3	3	2	2	2	-	-	1	-	-	2
CO3	2	3	2	2	3	-	-	1	1	-	2
CO4	3	2	2	2	3	-	-	1	1	-	3
Course Content											No of Lectures
Unit I: Information Theory Information Sources, Self-Information, Entropy, Joint Entropy, Conditional Entropy, Mutual Information, Source Coding Theorem, Shannon-Fano Coding, Huffman Coding, Run Length Encoding (RLE), Coding Efficiency.											[8]
Unit-II Channel Coding Discrete Memoryless Channel (DMC), Binary Symmetric Channel (BSC), Binary Erasure Channel (BEC), M-ary Symmetric Channel, M-ary Erasure Channel, Channel Capacity, Information Capacity Theorem, Shannon Limit, Channel Coding Theorem.											[8]
Unit-III Error Control Coding Error Detection and Correction Concepts, Hamming Weight and Hamming Distance, Minimum Distance Decoding, Parity Check Codes, Linear Block Codes, Generator Matrix, Hamming Codes, Syndrome Decoding, Cyclic Codes, Convolutional Codes, Trellis Representation, Viterbi											[8]

Approved by BoS of USAR : 16/6/2026

Approved by AC sub-committee : 20/07/2026

Applicable for Batch admitted in 2025 and Lateral Entry of AS 2026-27 admitted in Batch 2025

Decoding.	
Unit-IV Data Compression and IoT Applications Principles of Compression, Lossless Compression, Huffman Coding Applications, Arithmetic Coding, Run Length Encoding (RLE), Dictionary-Based Compression (LZW), Rate Distortion Function, Lossy Compression Fundamentals, Quantization, JPEG Lossless Compression, JPEG Lossy Compression, Video Compression Standards. Case Study: Role of AI in Data Compression and Error Control Coding for Resource-Constrained IoT Communication Systems.	[8]
Text Books [T1] Bose, R. (2016). Information theory, coding and cryptography (3rd ed.). McGraw-Hill Education. [T2] Cover, T. M., & Thomas, J. A. (2012). Elements of information theory (2nd ed.). Wiley.	
Reference Books [R1] Lin, S., & Costello, D. J. (2021). Error control coding (3rd ed.). Pearson. [R2] Sayood, K. (2023). Introduction to data compression (6th ed.). Morgan Kaufmann.	

Paper code: ARI473								L	P	Credit	
Paper Title: System on Chip								3	0	3	
Marking Scheme: Teachers Continuous Evaluation: As per university examination norms from time to time. End Term Theory Examination: As per university examination norms from time to time.											
INSTRUCTIONS TO PAPER SETTERS: Maximum Marks: As per University norms											
➤ There should be 9 questions in the end term examination question paper. ➤ Question No. 1 should be compulsory and cover the entire syllabus. This question should have objective or short answer type questions. ➤ Apart from Question No. 1, the rest of the paper shall consist of four units As per the syllabus. Every unit should have two questions. However, students may be asked to attempt only 1 question from each unit. ➤ The questions are to be framed keeping in view the learning outcomes of course/paper. The standard/ level of the questions to be asked should be at the level of the prescribed textbooks. ➤ The requirement of (scientific) calculators/ log-tables/ data-tables may be specified if required.											
Course Outcomes [Bloom’s Knowledge Level (KL)]: CO1: Explain SoC design methodologies, platform-based design approaches, multiprocessor SoCs, and low-power design concepts. [K2] CO2: Evaluate system models, validation and verification methods, and hardware/software co-design approaches for embedded systems. [K5] CO3: Evaluate communication-centric SoC design approaches and NoC architectures for optimized system performance. [K5] CO4: Design energy-efficient Network-on-Chip (NoC) architectures and protocols using low-power design techniques.											
CO/PO	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11
CO1	1	1	3	3	2	1	-	-	-	-	2
CO2	1	1	3	2	2	1	-	-	-	-	2
CO3	2	2	3	3	1	1	-	-	-	-	3
CO4	2	1	3	3	1	1	-	-	-	-	2
Course Content											No of Lectures
Unit-I Introduction To Soc Design: Typical Systems, Traditional Co-Design Methods, Tools Integrated Environment, Problem Statement, Platform-Based Soc Design Methodology: The Platform Concept, Platform-Centric Soc Design Approach, Comparison With Current Approaches, Other Embedded Design Approaches Using Uml, A Perspective On Collaboration With Non-Platform approaches, Multiprocessor Soc and Network on Chip, Low Power Soc Design.											[8]

Unit-II System Design with Model of Computation: System Models: Types of Models, Models of Computation, Validation and Verification, Hardware/Software Codesign Application Analysis, Synthesis.	[8]
Unit-III Computation–Communication Partitioning: Communication System: Current Trend, Separation of Communication and Computation, Communication Centric SoC Design, Communication Synthesis, Network-Based Design, Network on Chip, Architecture of NoC.	[8]
Unit-IV NOC Design: Practical Design of NoC, NoC topology-analysis Methodology, Energy Exploration, NoC Protocol Design, Low-Power Design For NoC: Low-Power Signaling, On-Chip Serialization, Low-Power Clocking, Low Power Channel Coding, Low-Power Switch, Low-Power Network On Chip Protocol.	[8]
Text Books [T1] Yoo, H. J., Lee, K., & Kim, J. K. (2018). Low-power noc for high-performance soc design (1st ed.). CRC press. [T2] Madiseti, V., & Arpnikanondt, C. (2006). A Platform-Centric Approach to System-on-Chip (SOC) Design (1st ed.). Springer Science & Business Media.	
Reference Books [R1] Martin, T. (2022). The designer's guide to the Cortex-M processor family (1st ed.). Newnes. [R2] Furber, S. B. (2000). ARM system-on-chip architecture (1st ed.). pearson Education.	

Paper code: ARI475							L	P	Credit		
Paper Title: Brain Computer Interface							3	0	3		
Marking Scheme:											
Teachers Continuous Evaluation: As per university examination norms from time to time.											
End Term Theory Examination: As per university examination norms from time to time.											
INSTRUCTIONS TO PAPER SETTERS: Maximum Marks : As per University norms											
➤ There should be 9 questions in the end term examination question paper. ➤ Question No. 1 should be compulsory and cover the entire syllabus. This question should have objective or short answer type questions. ➤ Apart from Question No. 1, the rest of the paper shall consist of four units As per the syllabus. Every unit should have two questions. However, students may be asked to attempt only 1 question from each unit. ➤ The questions are to be framed keeping in view the learning outcomes of course/paper. The standard/ level of the questions to be asked should be at the level of the prescribed textbooks. ➤ The requirement of (scientific) calculators/ log-tables/ data-tables may be specified if required.											
Course Outcomes [Bloom’s Knowledge Level (KL)]:											
CO1: Understand the fundamentals of Brain-Computer Interfaces, system design, and ethical aspects. [K1, K2]											
CO2: Learn key neural signal acquisition methods and brain potential patterns. [K2, K3]											
CO3: Gain knowledge of signal processing and feature extraction techniques in BCI. [K2, K3, K4]											
CO4: Apply machine learning approaches and explore practical applications of BCI. [K4, K5]											
CO/PO	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11
CO1	3	2	1	1	1	2	-	-	-	-	2
CO2	2	3	2	2	1	2	-	-	-	-	2
CO3	2	3	3	2	2	2	-	-	-	-	2
CO4	2	3	3	3	2	3	-	-	-	-	3
Course Content											No of Lectures
Unit-I Fundamentals of Brain-Computer Interfaces: Overview of brain structure and functions, Introduction to Brain-Computer Interfaces (BCI), Types of BCI: Synchronous and Asynchronous, Invasive, Partially Invasive, and Non-Invasive BCIs, Architecture of a BCI system, Monitoring hardware for BCI, Core techniques: EEG, ECoG, MEG, fMRI, Ethical considerations in Brain-Computer Interfacing.											[8]
Unit II Signal Acquisition and Neural Potentials: Brain activation patterns and brainwaves, Oscillatory potentials and event-related potentials, Error-related potentials and slow cortical potentials, Movement-related and stimulus-related potentials, Visual evoked potentials (including P300), Auditory evoked potentials, Potentials linked to cognitive tasks.											[8]

Unit III Signal Processing and Feature Extraction: Pre-processing of signals: filtering methods and artefact reduction, Frequency domain and time domain analysis, Wavelet-based analysis, Spatial filtering approaches, Introduction to Random Processes; Stationarity & Ergodicity, Auto- and cross-correlation functions.	[8]
Unit IV Machine Learning and Applications of BCI: Linear random processes; Auto-regressive, Moving average and ARMA models, Models for non-stationary processes; Trends, heteroskedasticity and ARIMA models, Fourier analysis of deterministic signals; Harmonic processes; SARIMA models, Evaluation metrics for BCI systems.	[8]
Text Books [T1] Rao, R. P. N. (2013). Brain-Computer Interfacing: An Introduction (1st ed.). Cambridge University Press. [T2] Wolpaw, J. R., & Wolpaw, E. W. (2012). Brain-Computer Interfaces: Principles and Practice (1st ed.). Oxford University Press.	
Reference Books [R1] Hassanien, A. E., & Azar, A. T. (Eds.). (2015). Brain-Computer Interfaces: Current Trends and Applications (1st ed.). Springer. [R2] Graimann, B., Allison, B., & Pfurtscheller, G. (Eds.). (2010). Brain-Computer Interfaces: Revolutionizing Human-Computer Interaction (1st ed.). Springer. [R3] Sanei, S., & Chambers, J. A. (2021). EEG Signal Processing and Machine Learning (2nd ed.). John Wiley & Sons.	

Paper code: ARI477T								L	P	Credits	
Subject: Image Processing and Computer Vision								3	0	3	
Marking Scheme: Teachers Continuous Evaluation: As per university examination norms from time to time. End Term Theory Examination: As per university examination norms from time to time.											
INSTRUCTIONS TO PAPER SETTERS: Maximum Marks: AS per University norms											
➤ There should be 9 questions in the end term examination question paper. ➤ Question No. 1 should be compulsory and cover the entire syllabus. This question should have objective or short answer type questions. ➤ Apart from Question No. 1, the rest of the paper shall consist of four units as per the syllabus. Every unit should have two questions. However, students may be asked to attempt only 1 question from each unit. ➤ The questions are to be framed keeping in view the learning outcomes of course/paper. The standard level of the questions to be asked should be at the level of the prescribed textbooks. ➤ The requirement of (scientific) calculators/ log-tables/ data-tables may be specified if required											
Course Outcomes [Bloom’s Knowledge Level (KL)]: Course Outcomes: CO1: Be able to formulate the mathematical operations for image processing. [K1, K2] CO2: Understand and apply the different filtering operations in spatial domain. [K3, K4] CO3: To Understand, apply and analyse image feature extraction and segmentation. [K2, K4] CO4: To Understand and apply concepts of 3D, camera matrix and motion analysis. [K3, K4]											
CO/PO	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11
CO1	3	3	3	2	3	1	-	2	-	1	1
CO2	3	2	3	2	3	1	-	2	-	1	1
CO3	3	2	3	3	3	1	-	2	-	1	1
CO4	3	2	3	3	3	1	-	2	-	1	1
Course Content											No of lectures
Unit I: Image Formation and Processing Introduction to Computer Vision, Photometric image formation, The digital camera, Geometric transformations, Image Enhancement: Point operators, neighborhood processing, Spatial filtering. Detection of Points, patches, Edge and Line, Canny edge detection.											[8]
Unit II: Interest Point, Region and Shape Representation Image interest point detection: interest points, aperture problem, Harries corner detection, Segmentation: Threshold based, Edge based and Region Based Segmentation, Boundary representation, Boundary Descriptors, Shape representation: Region based and contour based, Active contours (snakes).											[8]

Unit III: Image Understanding Image Understanding methods: bottom up and top down, Bottom-Up Approach: Image feature extraction (texture feature, SIFT, HOG), Machine learning/pattern recognition algorithms for object detection, object classification and segmentation. Top-down approach: Model based control, Active shape model, active appearance model, Point distribution models.	[8]
Unit IV: Camera Calibration and 3D Reconstruction Camera calibration, Camera parameter estimation, Triangulation, Two-frame structure from motion, Factorization, Parametric motion - Spline-based motion, Optical flow, Shape from X, Active range finding, Surface representations, Point-based representations, Volumetric representations, Model-based reconstruction.	[8]
Text Books: [T1] Szeliski, R. (2022). Computer Vision: Algorithms and Applications (2nd ed.). Springer. [T2] Gonzalez, R. C., & Woods, R. E. (2018). Digital Image Processing (4th ed.). Pearson.	
Reference Books: [R1] Trucco, E., & Verri, A. (1998). Introductory Techniques for 3-D Computer Vision (1st ed.). Prentice Hall. [R2] Treiber, M. (2010). An Introduction to Object Recognition: Selected Algorithms for a Wide Variety of Applications (1st ed.). Springer.	

Paper code: ARI479T										L	Credit
Paper Title: Generative Artificial Intelligence										3	3
Marking Scheme: Teachers Continuous Evaluation: As per university examination norms from time to time. End Term Theory Examination: As per university examination norms from time to time.											
INSTRUCTIONS TO PAPER SETTERS: Maximum Marks: As per University norms											
➤ There should be 9 questions in the end term examination question paper. ➤ Question No. 1 should be compulsory and cover the entire syllabus. This question should have objective or short answer type questions. ➤ Apart from Question No. 1, the rest of the paper shall consist of four units As per the syllabus. Every unit should have two questions. However, students may be asked to attempt only 1 question from each unit. ➤ The questions are to be framed keeping in view the learning outcomes of course/paper. The standard/ level of the questions to be asked should be at the level of the prescribed textbooks. ➤ The requirement of (scientific) calculators/ log-tables/ data-tables may be specified if required.											
Course Outcomes [Bloom's Knowledge Level (KL)]: CO1: Explain the fundamentals of Generative AI, representation learning, embeddings, transformers, and pre-trained language models. [K2] CO2: Describe the architecture and working of Large Language Models and image generative models, and analyze their applications. [K3, K4] CO3: Apply prompt engineering techniques and retrieval-based methods to effectively interact with and enhance generative AI systems. [K3] CO4: Identify tools, deployment concepts, ethical issues, and societal impacts associated with Generative AI systems. [K1, K2]											
CO \ PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	1	1	2	1	-	-	-	1	-
CO2	3	2	1	1	2	1	-	-	-	1	-
CO3	2	3	2	1	3	1	-	-	-	2	-
CO4	1	1	1	1	2	3	-	-	-	2	-
Course Content										No of Lectures	
Unit-I Introduction to Generative AI: Introduction to Generative AI and applications, Machine learning paradigms for generative models, Neural network basics and representation learning, Embeddings and language modelling concepts, Fundamentals of NLP: tokenization, Word2Vec and NER, Introduction to transformers and attention mechanism, Pre-trained language models: GPT and BERT.										[8]	
Unit-II Large Language Models: architecture overview and working, Zero-shot and few-shot learning, Popular LLMs: GPT, LLaMA and PaLM , Evaluation metrics for LLMs. Introduction to image generation: Autoencoders and GANs, Diffusion models: Stable Diffusion and DALL·E, Text-to-image generation and applications.										[8]	

Approved by BoS of USAR : 16/6/2026

Approved by AC sub-committee : 20/07/2026

Applicable for Batch admitted in 2025 and Lateral Entry of AS 2026-27 admitted in Batch 2025

Unit-III Prompt Engineering and Retrieval-Augmented Generation Prompt engineering fundamentals, Prompt design strategies and prompt patterns, Chain-of-thought prompting and self-consistency, Domain-specific prompting, Introduction to vector embeddings and semantic search, Vector databases, Retrieval-Augmented Generation (RAG) architecture and use cases.	[8]
Unit-IV Generative AI frameworks and tools: Hugging Face and PyTorch, LangChain basics, Applications of Generative AI in education, healthcare and software development, Introduction to model deployment concepts. Ethics and risks in Generative AI: bias, hallucination, deepfakes and privacy, Responsible and trustworthy AI.	[8]
Text Books [T1] Baker, P. (2024). Generative AI for dummies (1st ed.). John Wiley & Sons. [T2] Foster, D. (2022). Generative deep learning (2nd ed.). " O'Reilly Media, Inc." [T3] Goodfellow, I., Bengio, Y., Courville, A., & Bengio, Y. (2016). Deep learning (1st ed.). Cambridge: MIT press.	
Reference Books [R1] Russel, S., & Norvig, P. (2015). “Artificial Intelligence–A Modern Approach” (4th Global ed.). Pearson Education, 2003. [R2] Bok, V., & Langr, J. (2019). GANs in action: deep learning with generative adversarial networks (1st ed.). Simon and Schuster.	